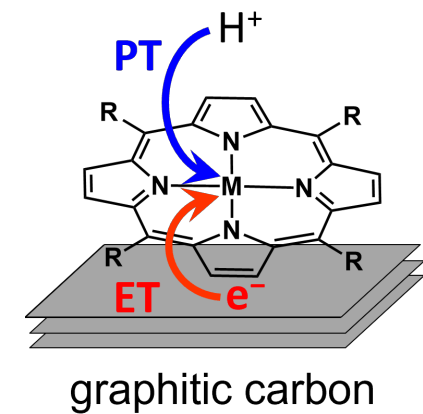
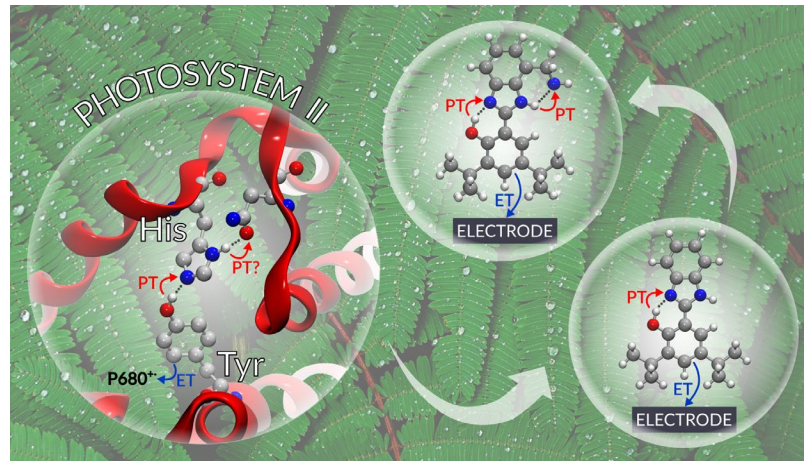
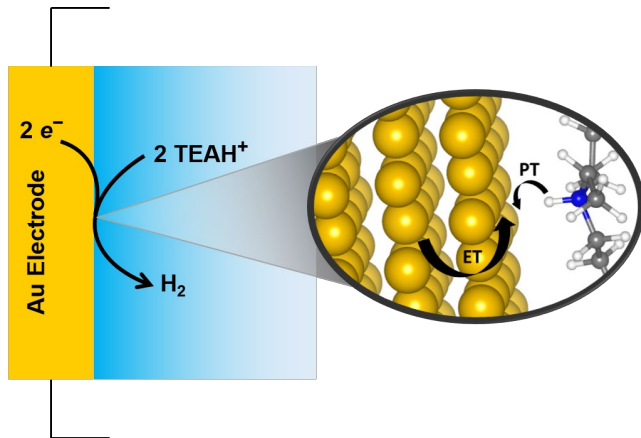


Modeling Proton-Coupled Electron Transfer in Solution and at Interfaces

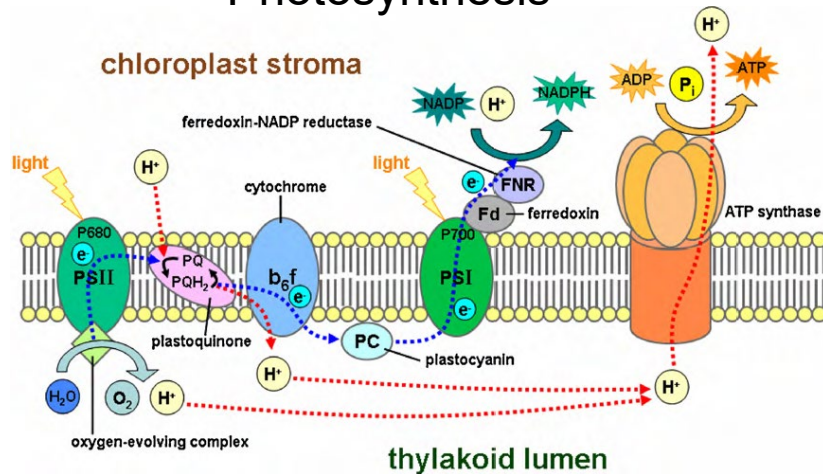
Sharon Hammes-Schiffer
Princeton University



Proton-Coupled Electron Transfer (PCET)

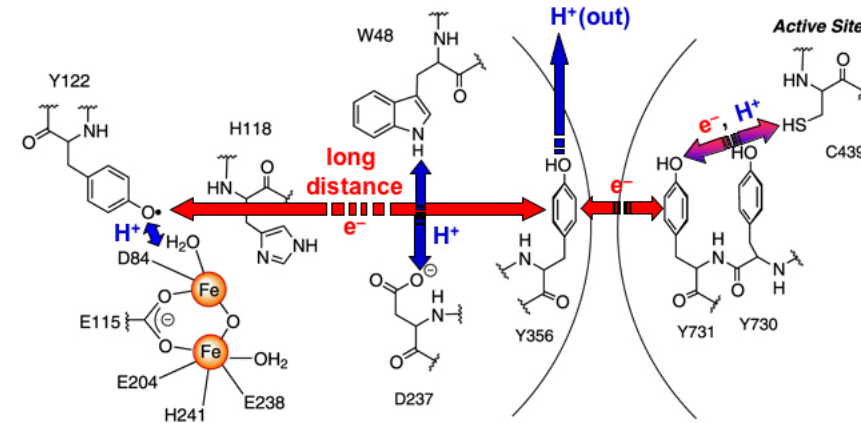
- **Biological processes:** photosynthesis, respiration, enzymes
- **Chemical processes:** fuel cells, solar cells, energy devices

Photosynthesis

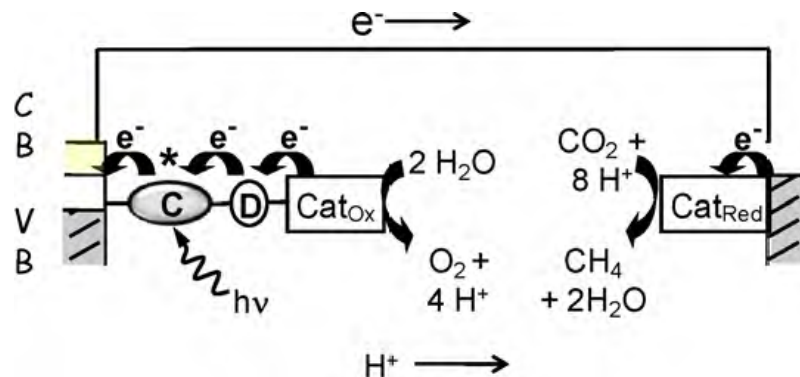


<http://photosynthesiseducation.com/photosynthesis-in-plants/>

Class I Ribonucleotide Reductase



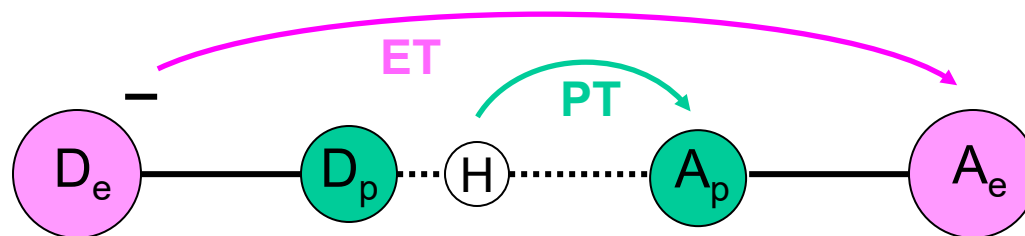
Stubbe group: <http://web.mit.edu/biochemistry/research.htm>



Dye sensitized photoelectrochemical cell

Gagliardi, Westlake, Kent, Paul, Papanikolas, Meyer, *CCR* **254**, 2459 (2010).

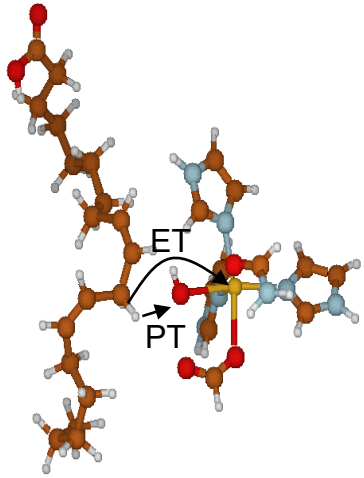
Theory of Proton-Coupled Electron Transfer



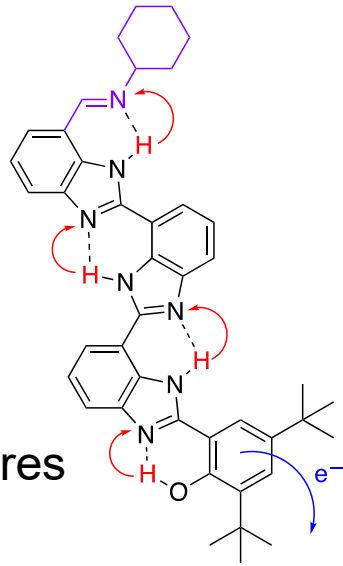
- Proton-coupled electron transfer (PCET)
 - electron & proton transfer between different or same donors/acceptors
 - electron & proton transfer in the same or different directions
 - sequential or concerted mechanisms
 - potentially multi-electron and multi-proton transfers
- PCET theory*
 - Quantum mechanical effects of electrons and transferring proton(s)
 - Coupling and varying timescales among electrons, protons, donor-acceptor modes, solvent, protein

*Soudackov and Hammes-Schiffer, *JCP* 1999, 2000, 2005; *JPC B* 2008.
Hammes-Schiffer and Stuchebrukhov, *Chem. Rev.* 2010.

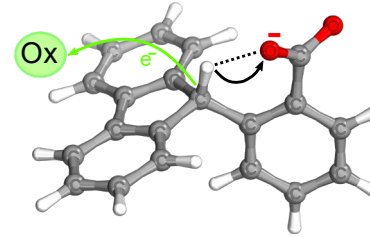
Applications of Theory to PCET



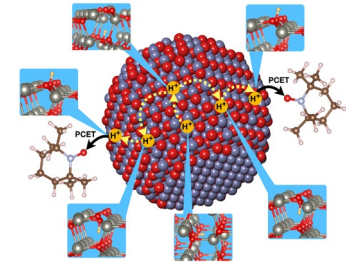
Enzyme: large kinetic isotope effects k_H/k_D up to 700



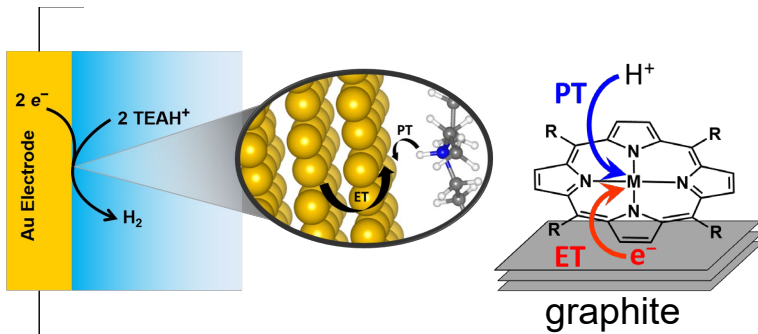
Proton wires



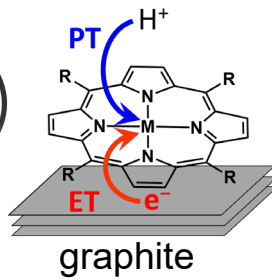
CH bond activation



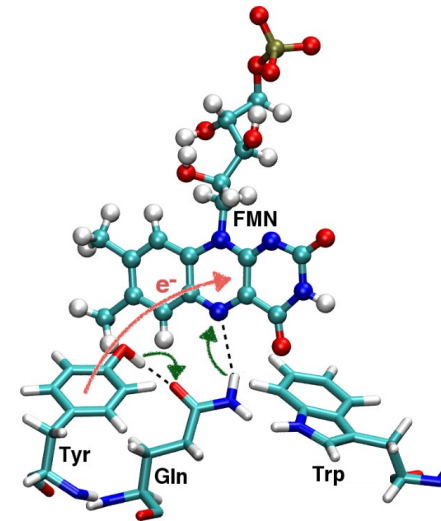
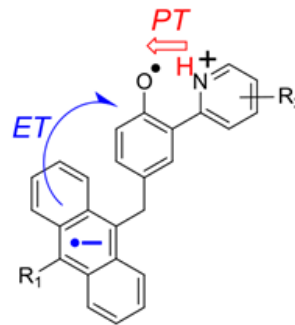
Photoreduced ZnO nanocrystals



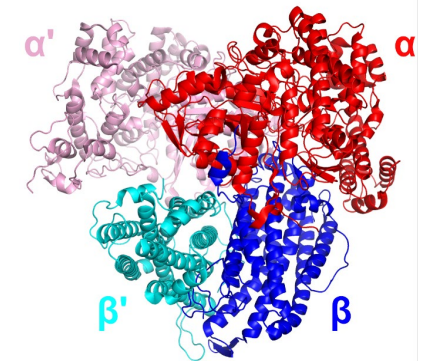
Electrochemical PCET



Inverted region for PCET and PCEnT

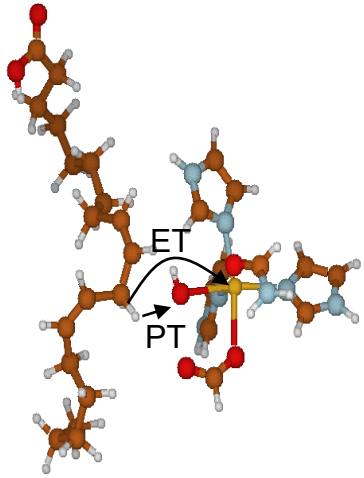


BLUF photoreceptor

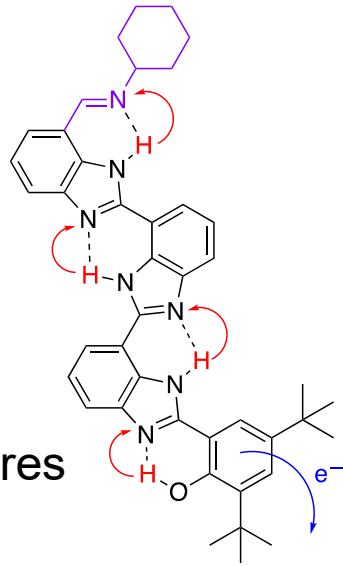


Ribonucleotide reductase: 6 PCET reactions over 32 Å

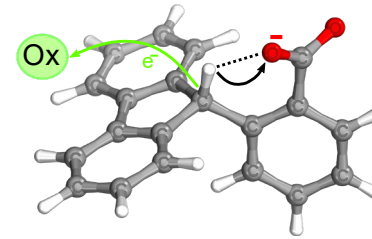
Applications of Theory to PCET



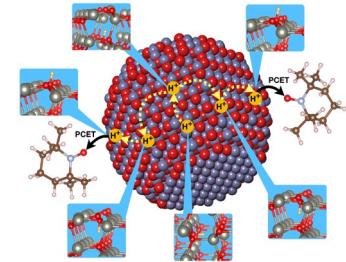
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 k_H/k_D up to 700



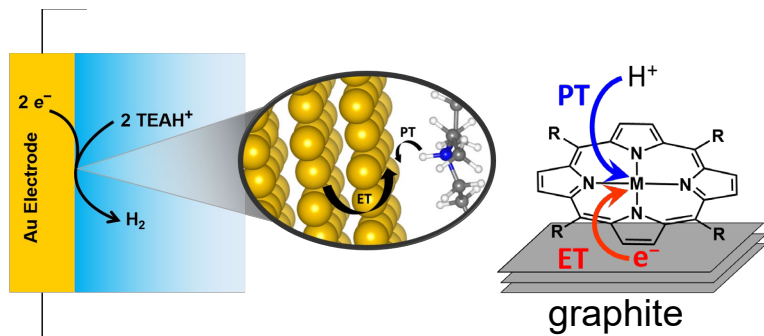
Proton wires



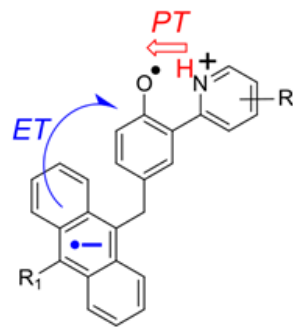
CH bond activation



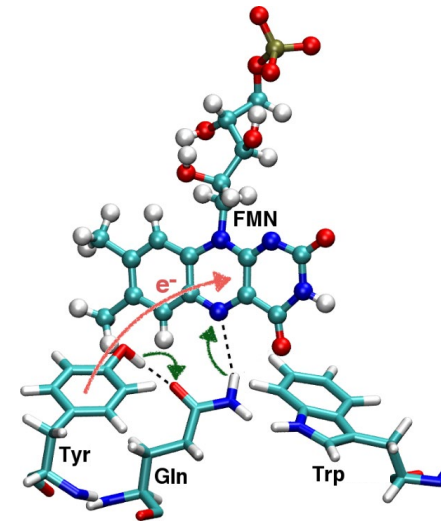
Photoreduced ZnO nanocrystals



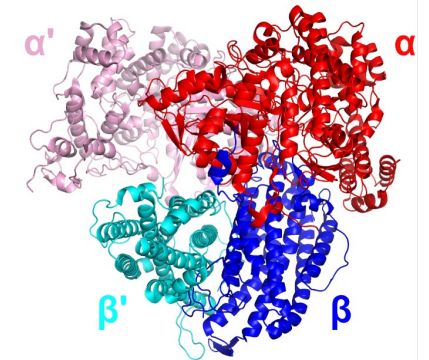
Electrochemical PCET



Inverted region for PCET and PCEnT



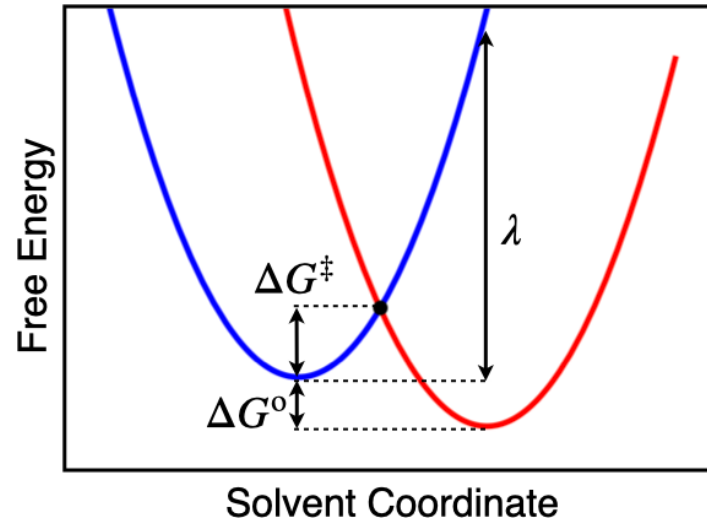
BLUF photoreceptor



Ribonucleotide reductase:
6 PCET reactions over 32 Å

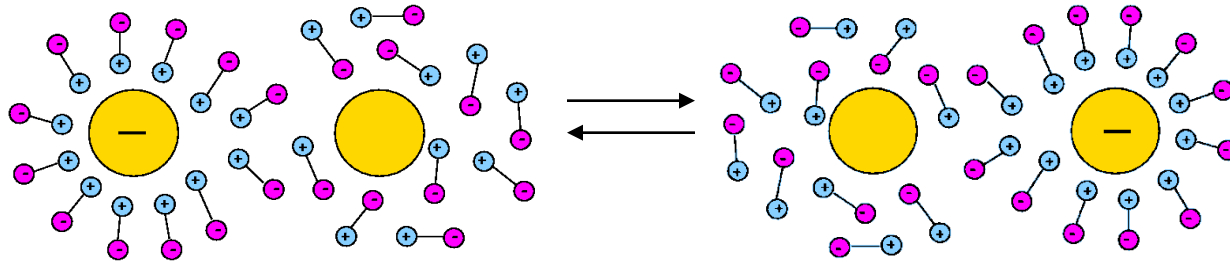
Single Electron Transfer

Diabatic states:



Marcus theory

Other major contributors:
Hush, Levich, Jortner, ...



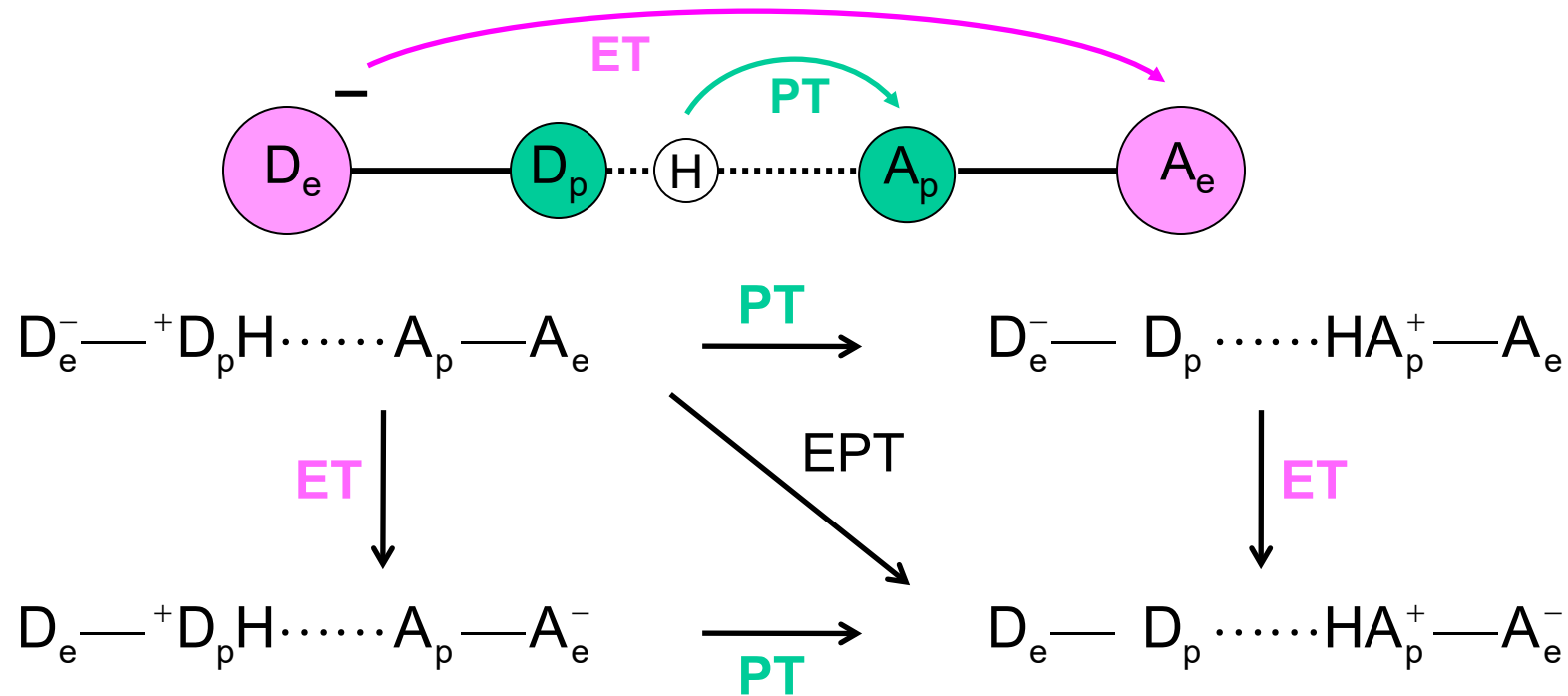
Nonadiabatic ET rate constant:
$$k = \frac{2\pi}{\hbar} V_{12}^2 (4\pi\lambda k_B T)^{-1/2} \exp\left[-\Delta G^\ddagger / (k_B T)\right]$$

Concepts extended to PT:
Hynes and coworkers
ET&PT work: Warshel

$$\Delta G^\ddagger = (\Delta G^\circ + \lambda)^2 / (4\lambda)$$

V_{12} : coupling between diabatic states

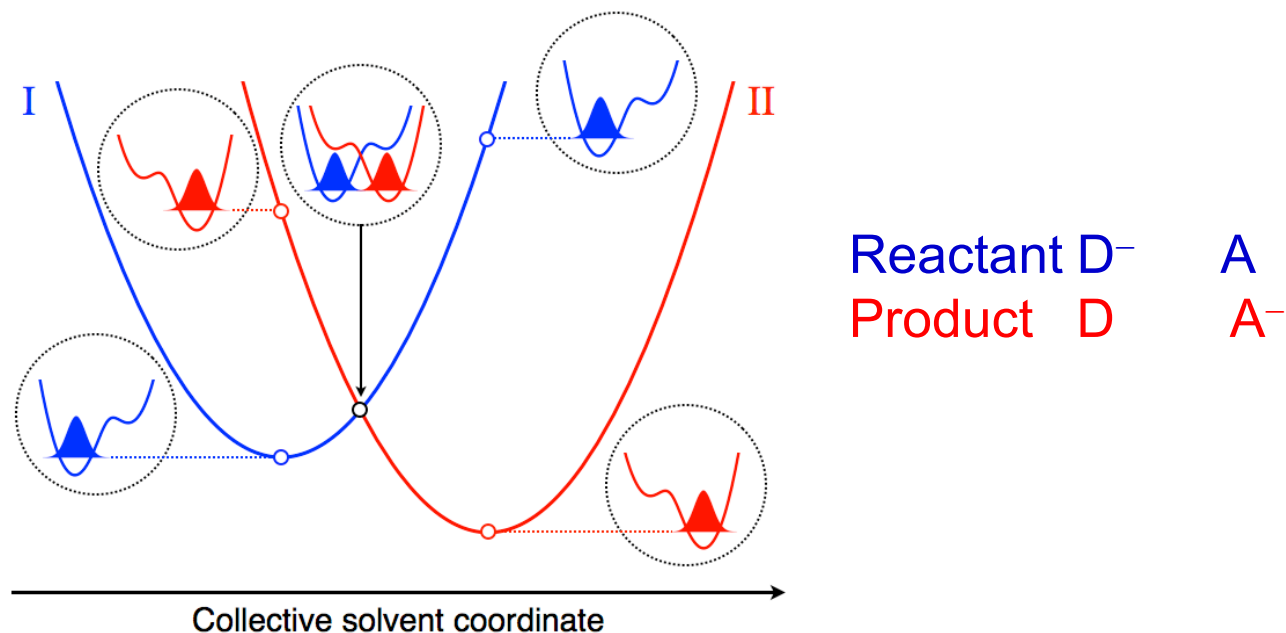
Proton-Coupled Electron Transfer Theory



- Sequential: ET-PT or PT-ET
- Concerted: EPT
- Mechanism determined by relative energies and couplings
- Off-diagonal states much higher in energy → concerted EPT

Concerted PCET (EPT) Theory

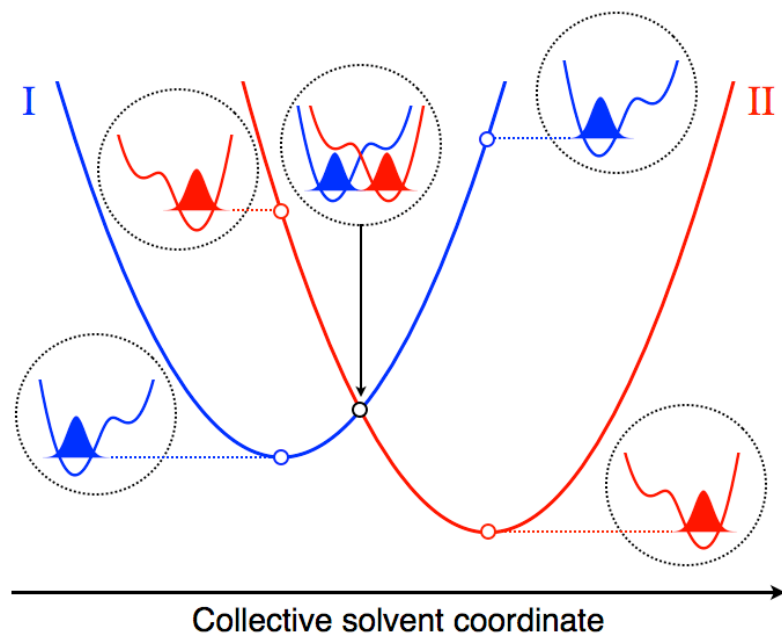
Soudackov and Hammes-Schiffer, JCP 111, 4672 (1999)



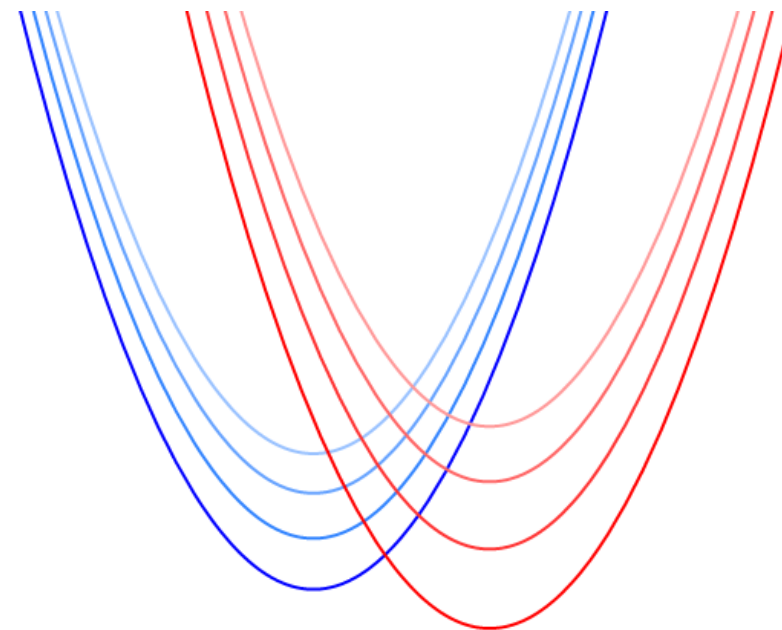
- Free energy curves are functions of collective solvent coordinate
- H nucleus: quantum mechanical wavefunction
- Electron-proton vibronic states corresponding to different proton vibrational states for each electronic state

Concerted PCET (EPT) Theory

Soudackov and Hammes-Schiffer, JCP 111, 4672 (1999)

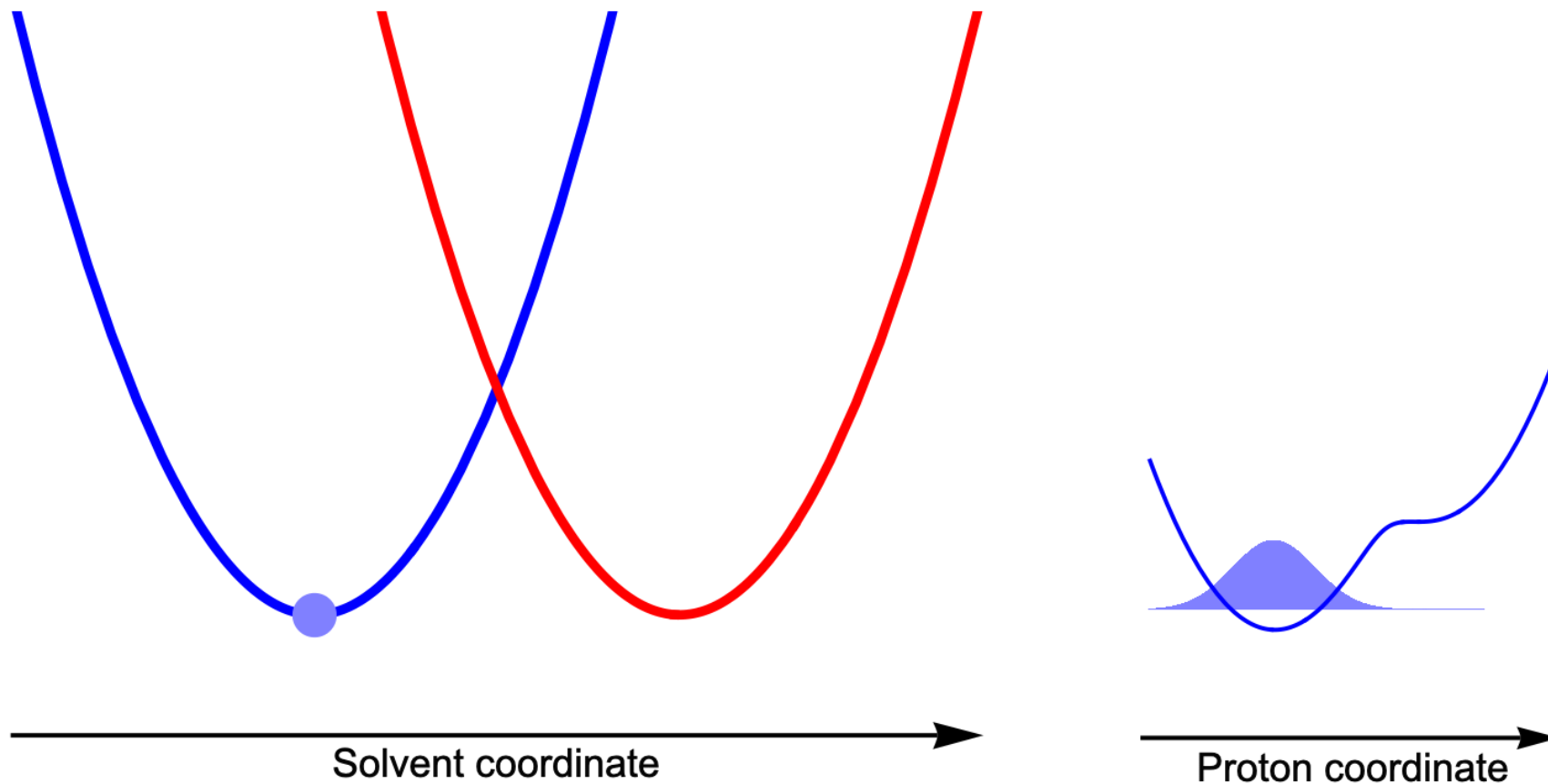


Reactant D^- A
Product D A^-

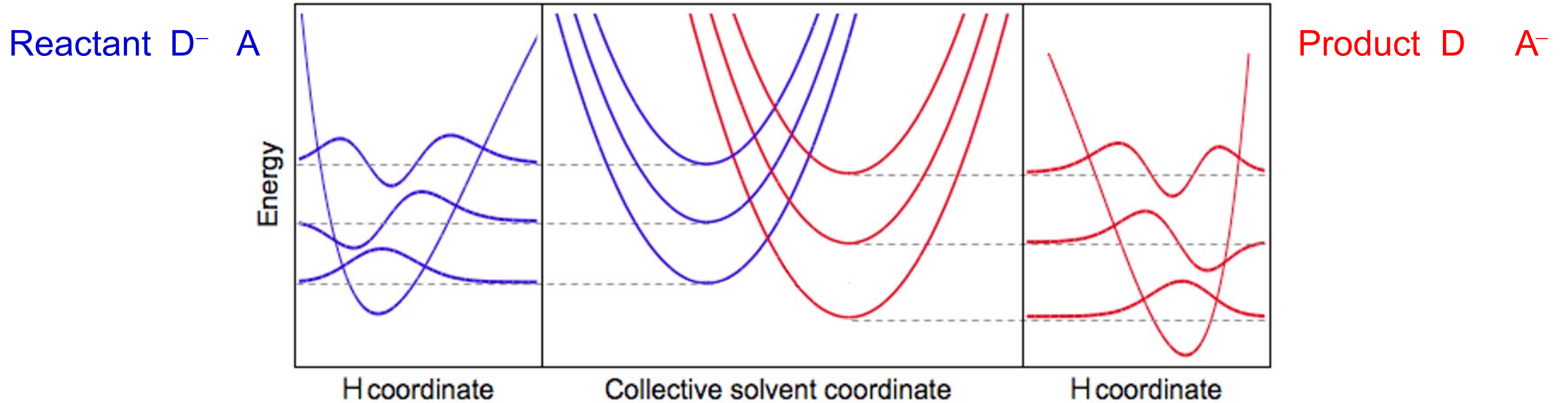


- Free energy curves are functions of collective solvent coordinate
- H nucleus: quantum mechanical wavefunction
- Electron-proton vibronic states corresponding to different proton vibrational states for each electronic state

Fundamental Mechanism for PCET

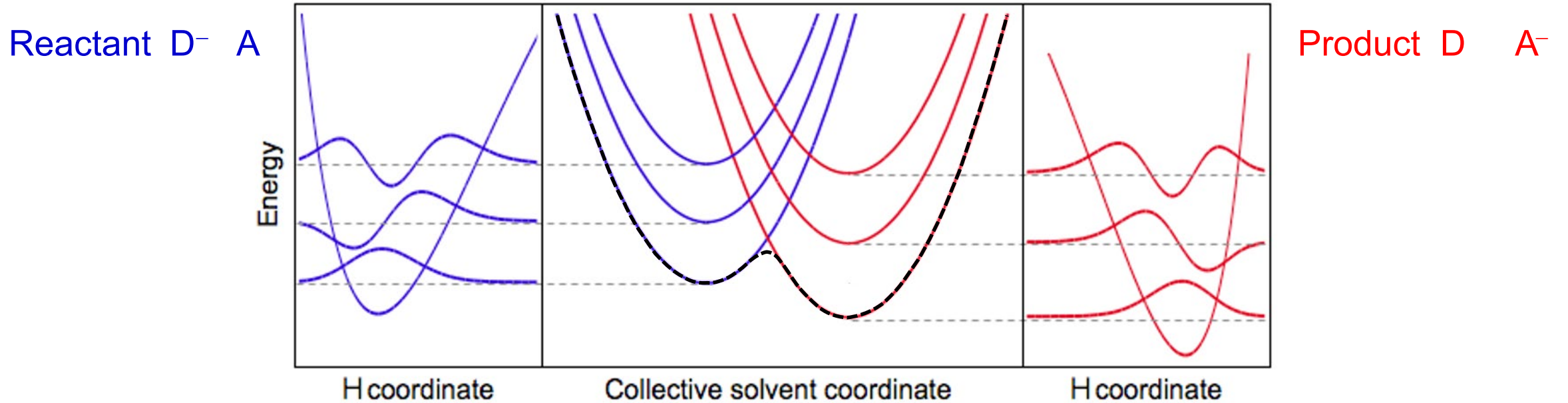


Two Sets of Vibronic States



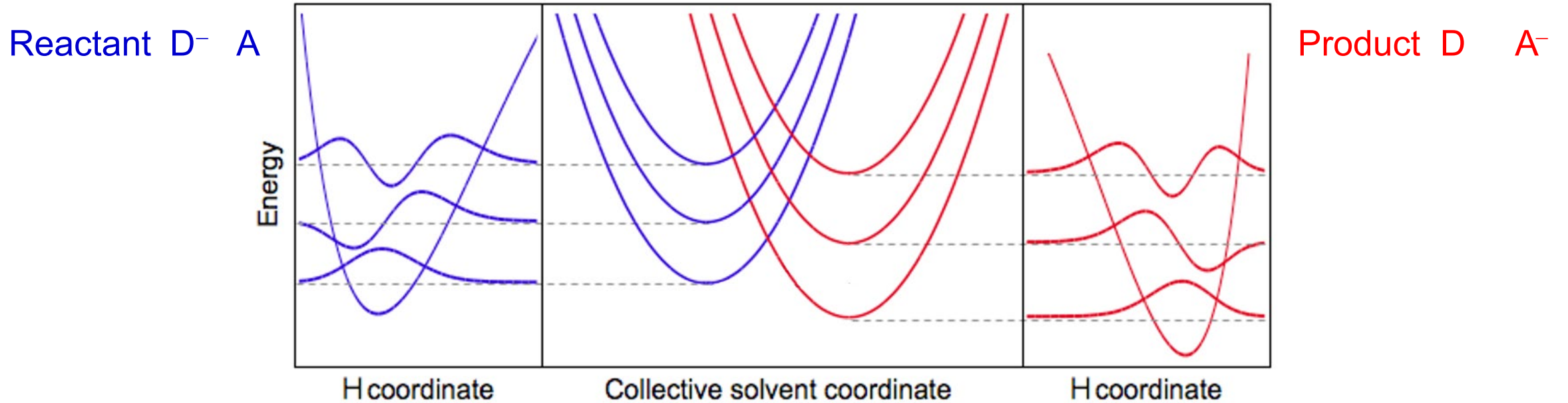
- Vibronic states correspond to different proton vibrational states for each electronic state
- Vibronically **adiabatic**: PCET occurs on **ground** vibronic state (transition state theory)
- Vibronically **nonadiabatic**: PCET involves **excited** vibronic states
 - nonadiabatic transitions between pairs of reactant and product vibronic states (golden rule)

Two Sets of Vibronic States



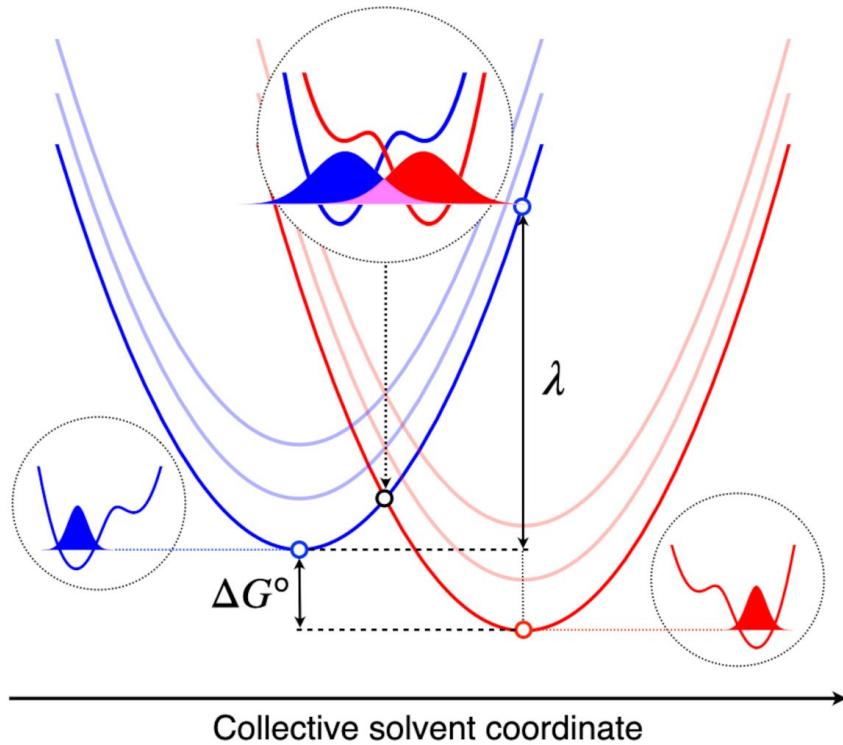
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Two Sets of Vibronic States



- Vibronic states correspond to different proton vibrational states for each electronic state
- Vibronically **adiabatic**: PCET occurs on **ground** vibronic state (transition state theory)
- Vibronically **nonadiabatic**: PCET involves **excited** vibronic states
 - nonadiabatic transitions between pairs of reactant and product vibronic states (golden rule)

PCET Rate Constant Expressions

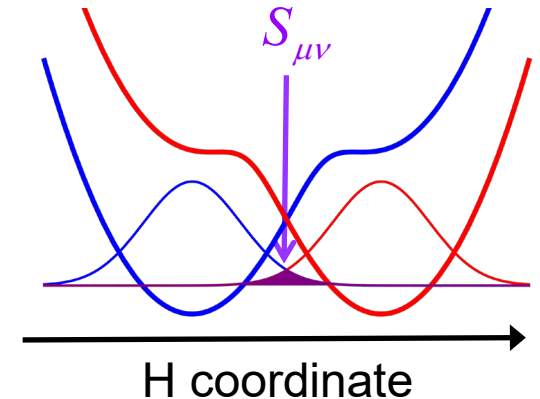


Most PCET reactions are nonadiabatic (involve excited vibronic states) due to small vibronic coupling

$$k = \frac{2\pi}{\hbar} \sum_{\mu} P_{\mu} \sum_{\nu} (4\pi\lambda k_B T)^{-1/2} (V_{\mu\nu})^2 \exp\left[-\Delta G_{\mu\nu}^{\ddagger} / (k_B T)\right]$$

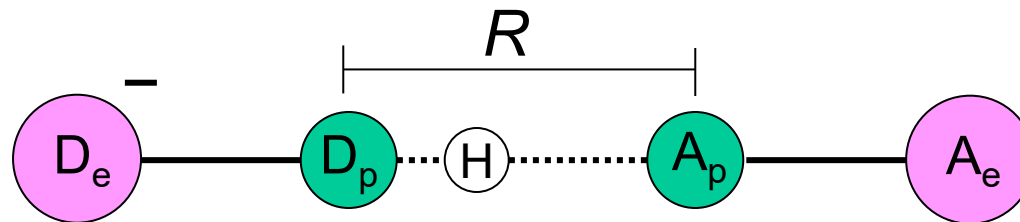
$$\Delta G_{\mu\nu}^{\ddagger} = (\Delta G_{\mu\nu}^{\circ} + \lambda)^2 / (4\lambda)$$

$$V_{\mu\nu} = V^{\text{el}} S_{\mu\nu}$$

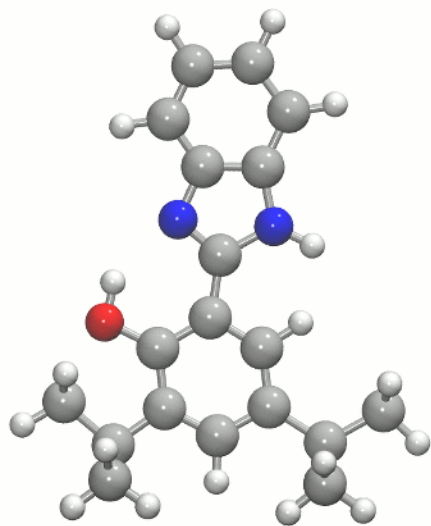


Form of vibronic coupling depends on electron-proton nonadiabaticity

Include Proton Donor-Acceptor Motion



Derived analytical rate constant expressions in various regimes,
approximating overlap as decreasing exponentially with R



R is O—N distance

Alternative (often equivalent) method:
Thermal averaging over R via probability distribution $P(R)$:

$$k = \int_0^{\infty} k(R)P(R)dR$$

Analytical derivations:

Soudackov, Hatcher, SHS, JCP 2005

Soudackov, SHS, JCP 2015, Faraday Discuss. 2016

Input Quantities for PCET Rate Constants

- Reaction free energy, ΔG°
- Electronic coupling, V^{el}
- Reorganization energy, λ
- Proton potential energy curves $\rightarrow$ proton vibrational wavefunctions and energy levels
- Probability distribution function for proton donor acceptor distance R , $P(R)$

$$k(R) = \frac{2\pi}{\hbar} (V^{\text{el}})^2 \sum_{\mu} P_{\mu} \sum_{\nu} (4\pi\lambda k_B T)^{-1/2} (S_{\mu\nu})^2 \exp\left[-(\Delta G^\circ + \Delta\epsilon_{\mu\nu} + \lambda)^2 / (4\lambda k_B T)\right]$$
$$k^{\text{PCET}} = \int dR k(R) P(R)$$

Reaction Free Energy and Electronic Coupling

- Reaction free energy, ΔG°
- Electronic coupling, V^{el}
- Reorganization energy, λ
- Proton potential energy curves $\rightarrow$ proton vibrational wavefunctions and energy levels
- Probability distribution function for proton donor acceptor distance R , $P(R)$

ΔG° : experiment, DFT calculations, QM/MM free energy simulations

V^{el} : constrained DFT, CASSCF, not needed for relative rate constants and KIEs

λ : sum of inner-sphere (solute) and outer-sphere (solvent) contributions

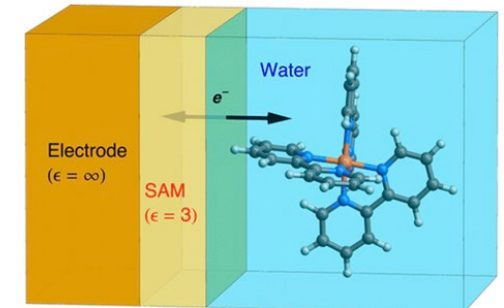
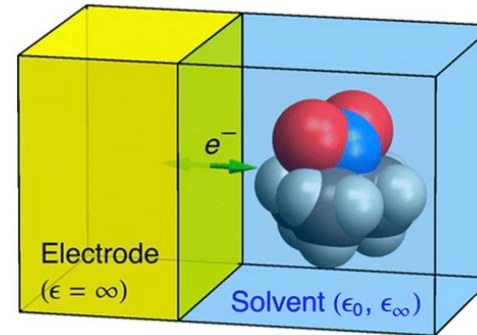
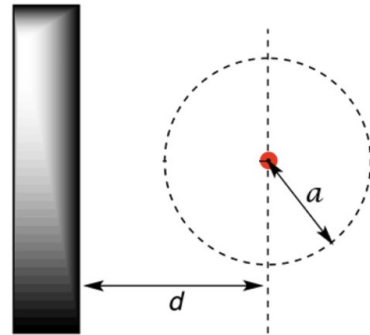
Reorganization Energy

- Total λ is sum of inner-sphere (solute) and outer- sphere (solvent) reorganization energies
- Inner-sphere: four-point scheme, optimize geometries of oxidized and reduced states

$$\lambda_i = \frac{1}{2} [E_{\text{Ox}}(Q^{\text{Red}}) - E_{\text{Ox}}(Q^{\text{Ox}}) + E_{\text{Red}}(Q^{\text{Ox}}) - E_{\text{Red}}(Q^{\text{Red}})]$$

- Outer-sphere: analytical expressions, more elaborate dielectric continuum models, molecular dynamics simulations

$$\lambda_o = \frac{(\Delta q)^2}{2} \left(\frac{1}{\epsilon_\infty} - \frac{1}{\epsilon_0} \right) \left(\frac{1}{a} - \frac{1}{2d} \right)$$



Nelsen, Blackstock, and Kim, JACS **109**, 677 (1987).

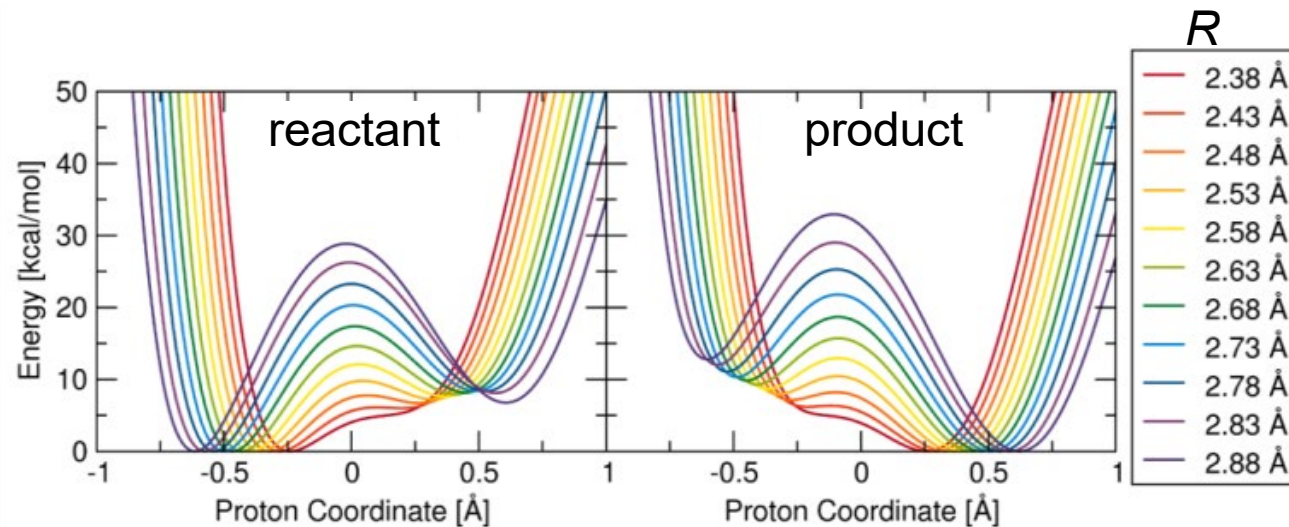
Auer, Fernandez, and Hammes-Schiffer, JACS **133**, 8282 (2011).

Liu and Newton, JPC **98**, 7162 (1994).

Ghosh, Horvath, Soudackov, Hammes-Schiffer, JCTC **10**, 2091 (2014).

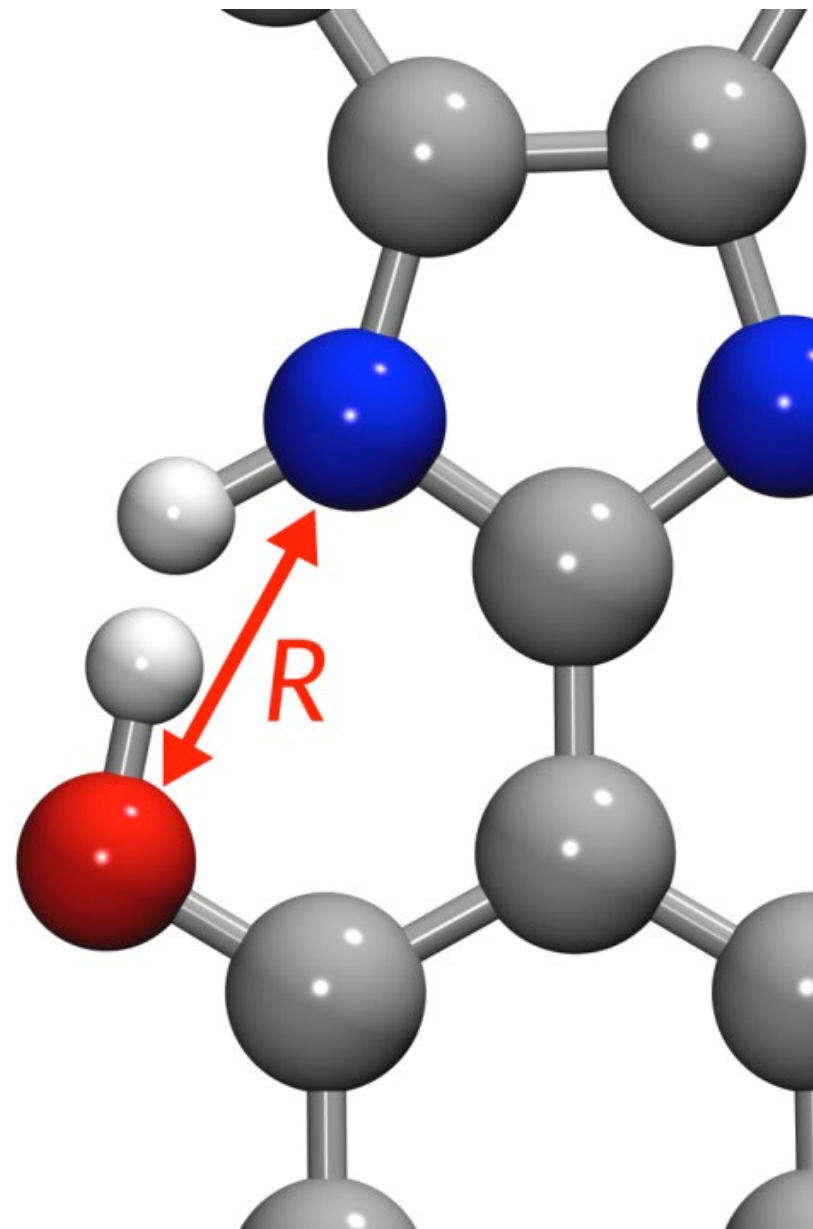
Proton Potential Energy Curves

- Reaction free energy, ΔG°
- Electronic coupling, V^{el}
- Reorganization energy, λ
- Proton potential energy curves $\rightarrow$ proton vibrational wavefunctions and energy levels
- Probability distribution function for proton donor acceptor distance R , $P(R)$
 - Compute average reactant and product geometries for each R
 - Compute proton potential energy curves for each average geometry



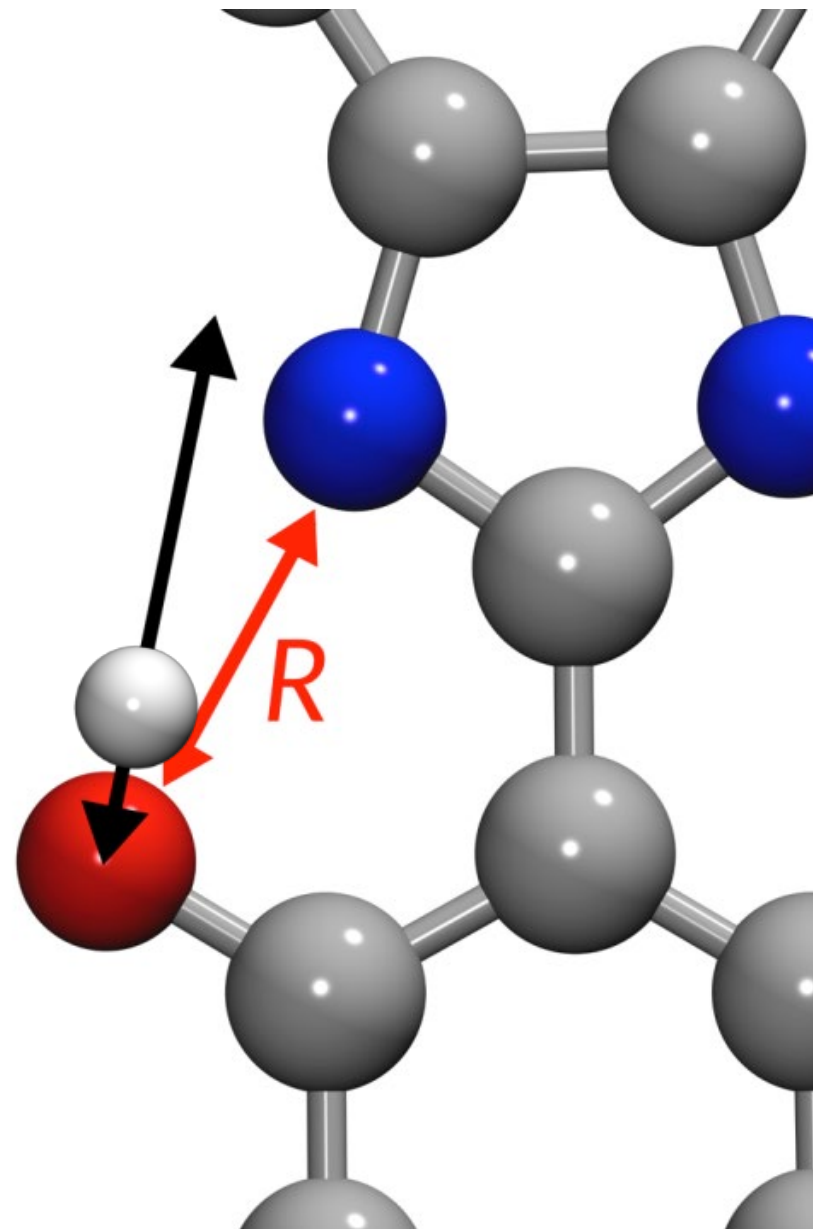
Generate Proton Potentials

1. Define proton coordinate using proton positions.
2. Move proton along one-dimensional grid spanning axis.



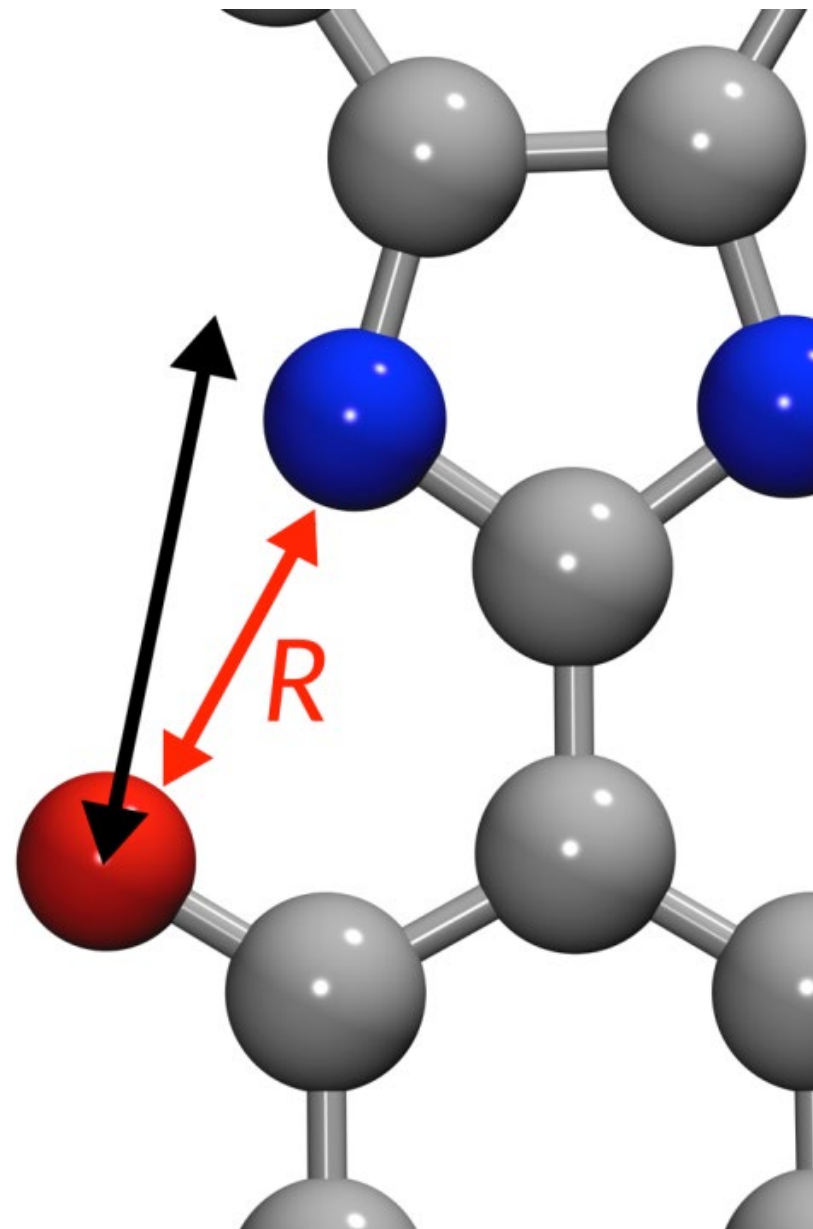
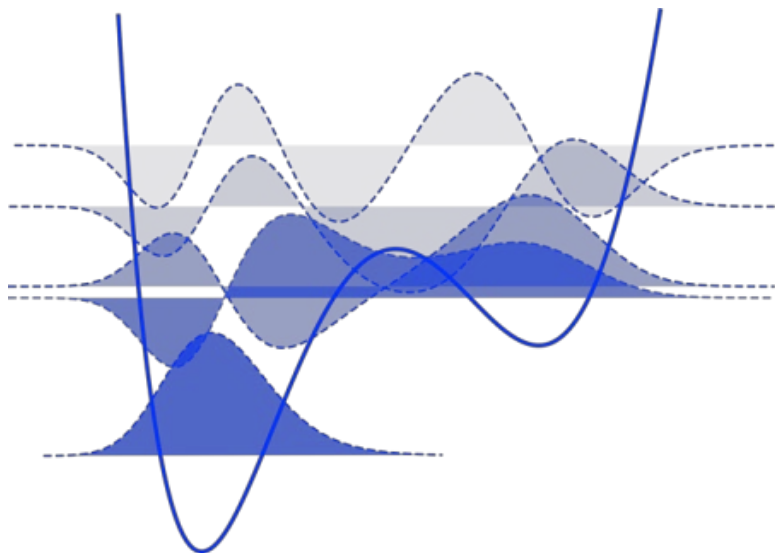
Generate Proton Potentials

1. Define proton coordinate using proton positions.
2. Move proton along one-dimensional grid spanning axis.
3. Calculate single point energies along grid points.
4. Calculate proton vibrational wave functions for each proton potential (fixed R).



Generate Proton Potentials

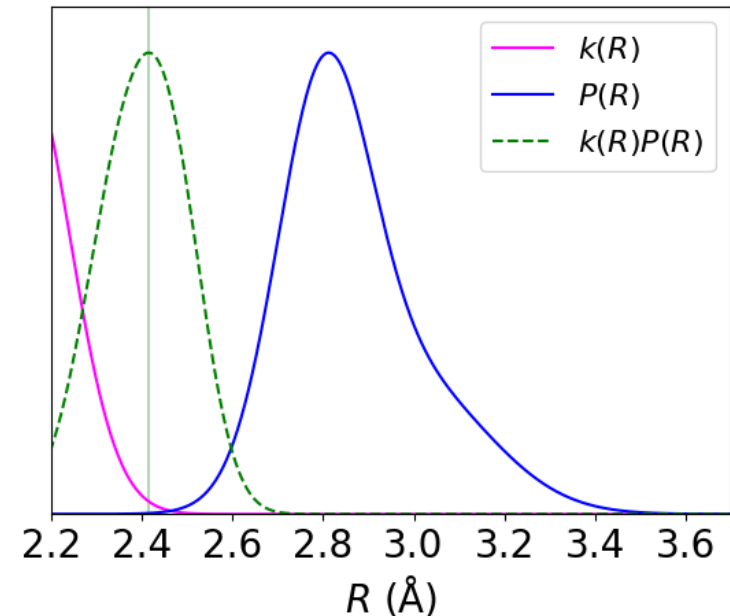
1. Define proton coordinate using proton positions.
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4. Calculate proton vibrational wave functions for each proton potential (fixed R).



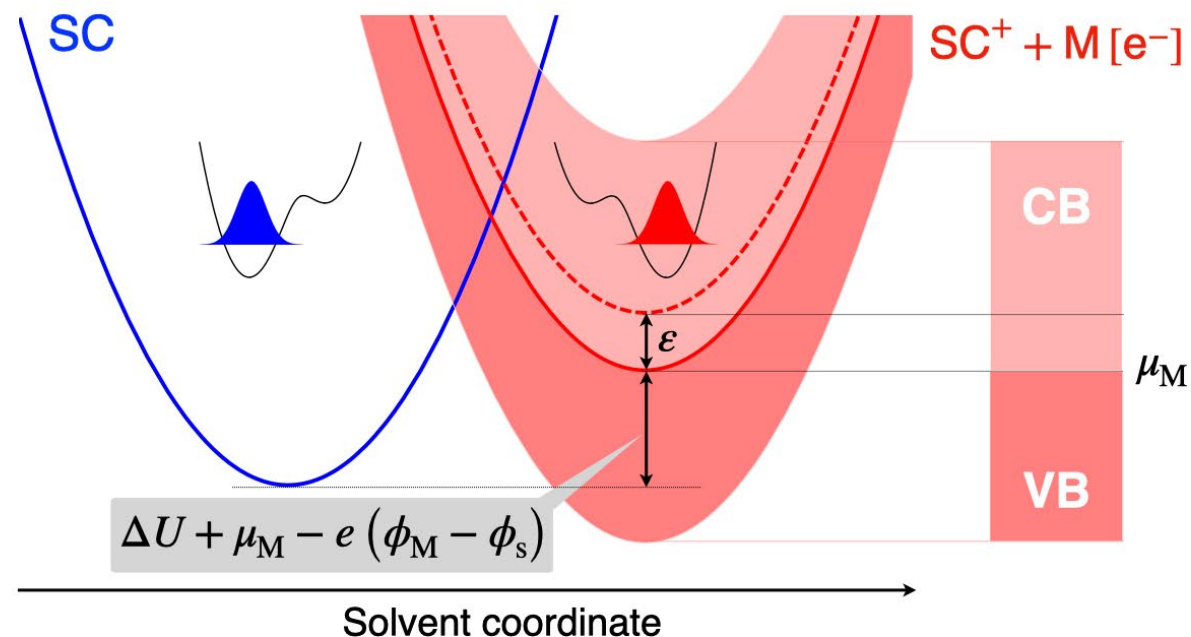
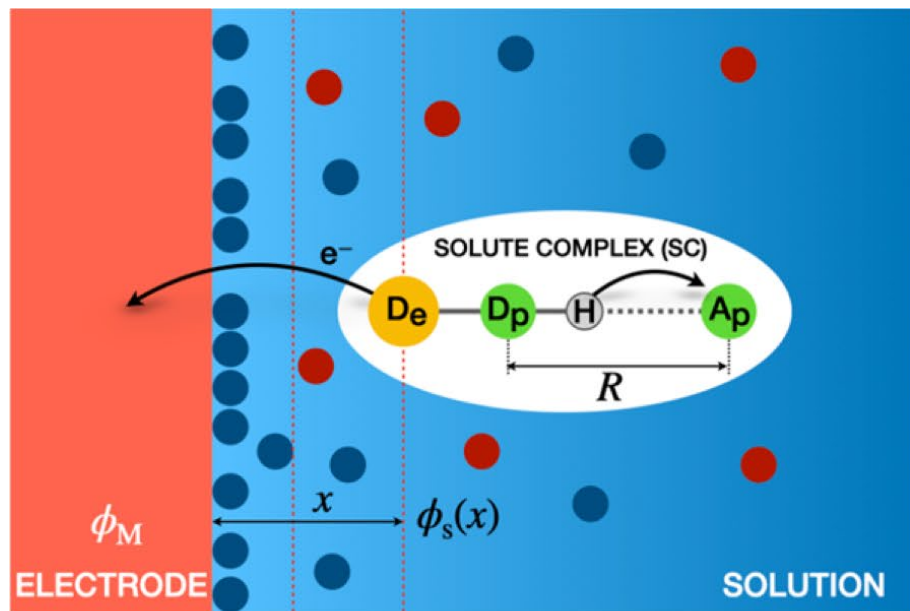
Input Quantities for PCET Rate Constants

- Reaction free energy, ΔG°
 - Electronic coupling, V^{el}
 - Reorganization energy, λ
 - Proton potential energy curves $\rightarrow$ proton vibrational wavefunctions and energy levels
 - Probability distribution function for proton donor acceptor distance R , $P(R)$
-
- DFT energies from geometry optimizations of reactant with constrained R
 - Molecular dynamics simulations using umbrella sampling

$$k^{\text{PCET}} = \int dR \, k(R) P(R)$$



Electrochemical PCET Theory

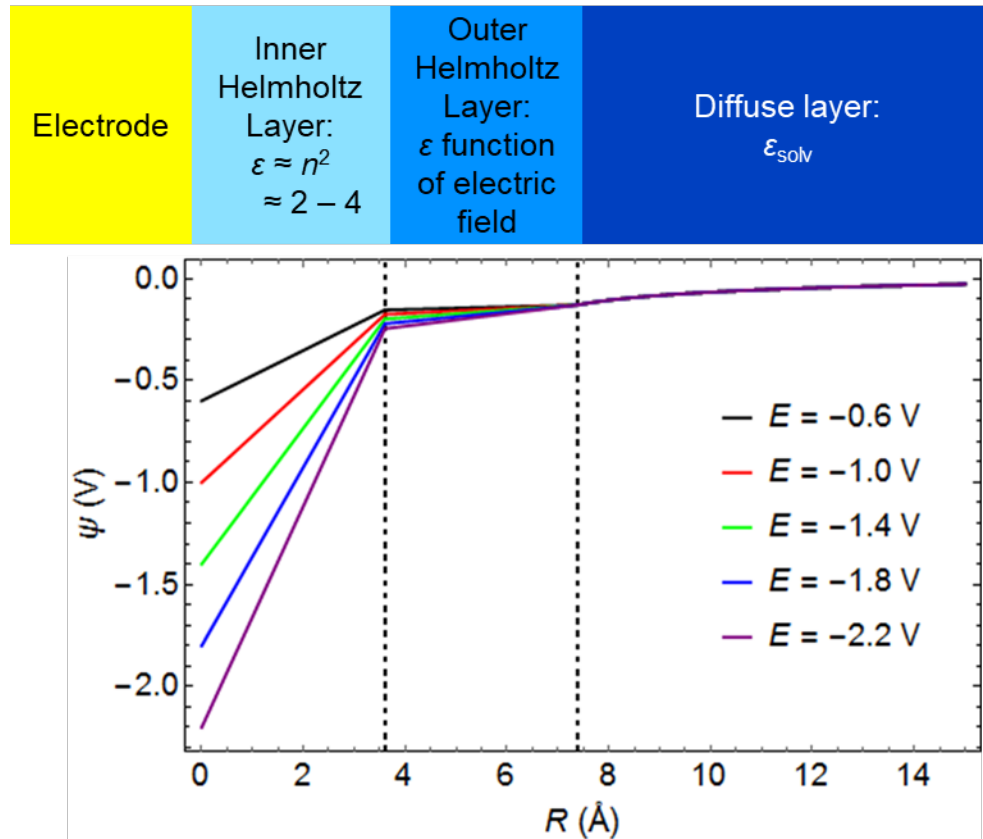
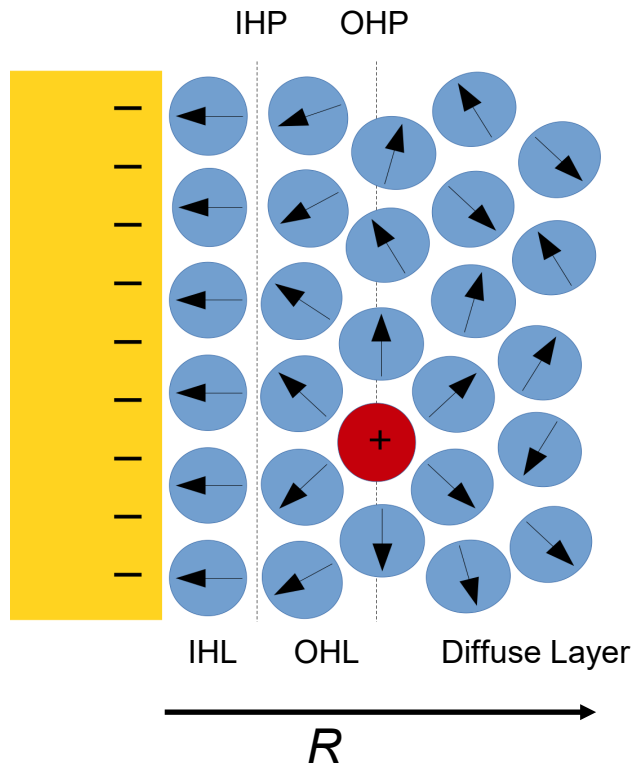


Vibronically nonadiabatic PCET theory extended to electrochemical PCET

- Continuum of electronic states in metal electrode
- ΔG depends on applied potential
- Electrical double layer

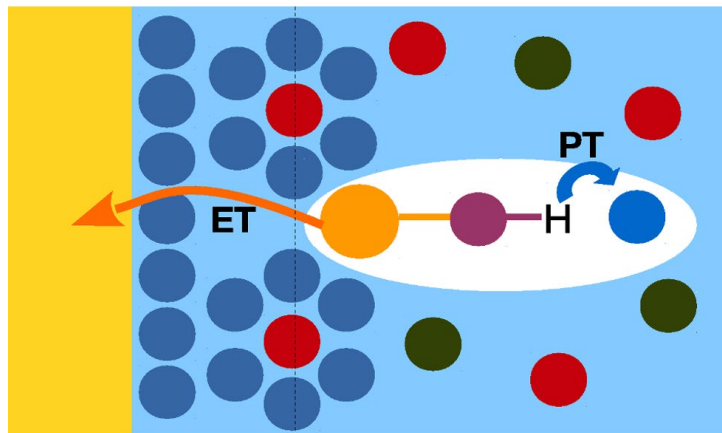
Venkataraman, Soudackov, and Hammes-Schiffer, JPC C 2008.

Electrical Double Layer Model

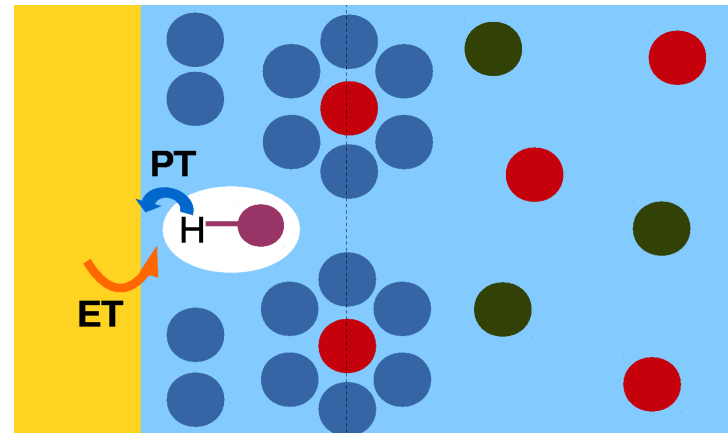


- Extended Guoy-Chapman-Stern model: multilayer dielectric continuum
- Solve Laplace and Langevin-Poisson-Boltzmann equations self-consistently for electrode potential E
- Electrostatic potential increases linearly in IHL and OHL
- Include $e\psi(R)$ in ΔG

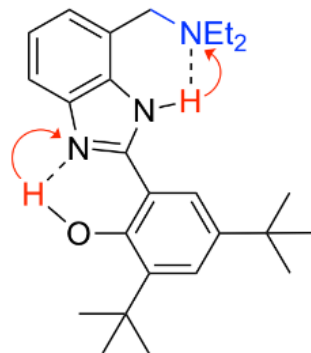
Homogeneous vs Heterogeneous PCET



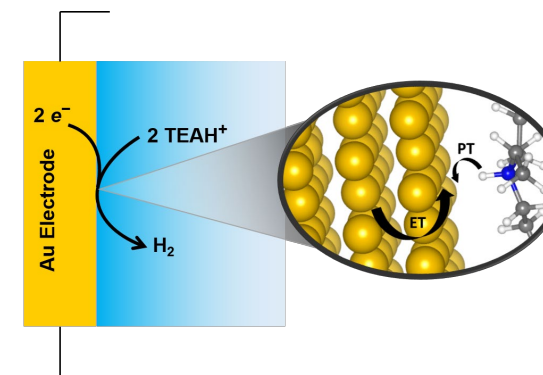
Homogeneous electrochemical PCET



Heterogeneous electrochemical PCET

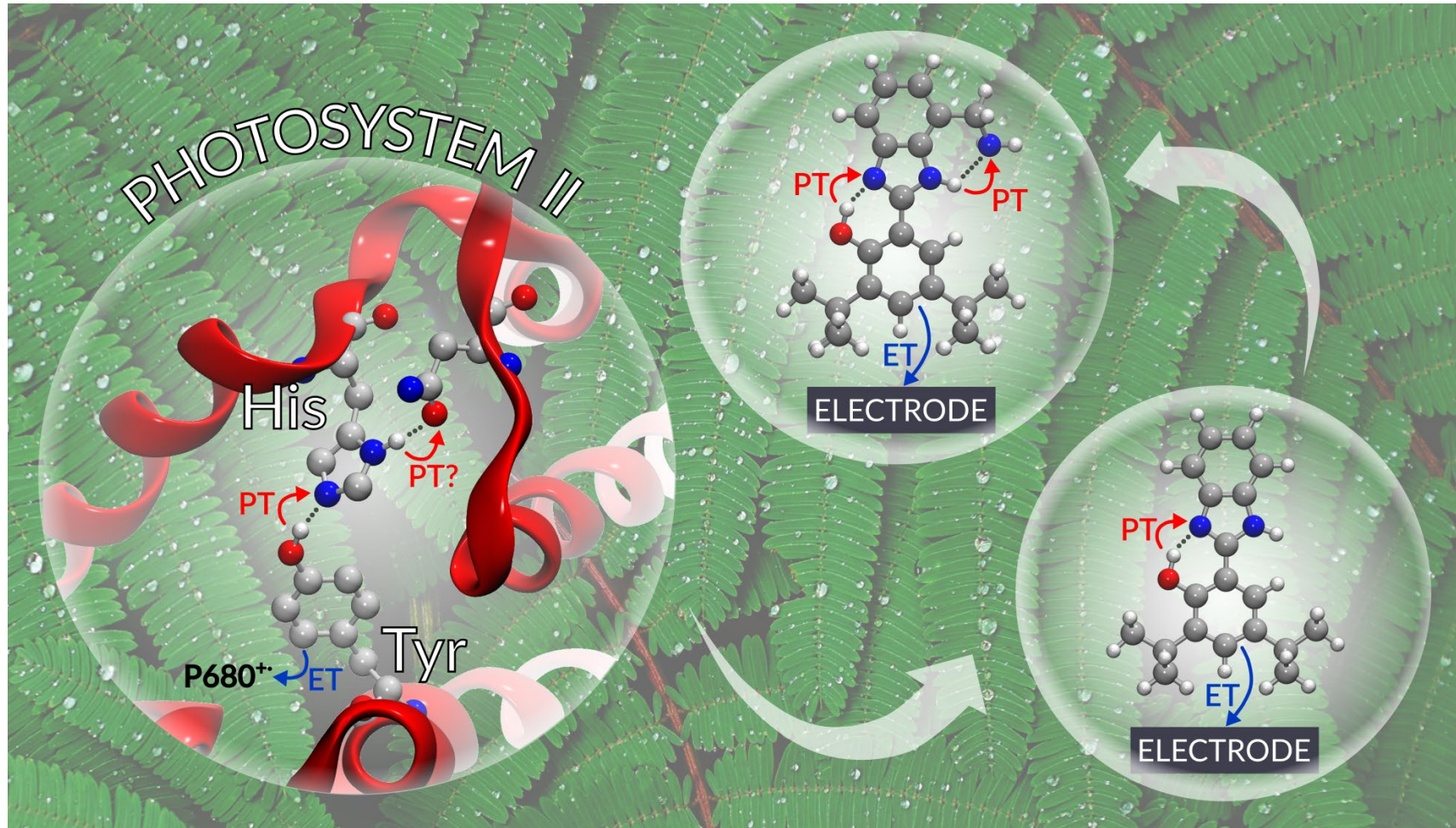


proton wires



proton discharge to
electrode surface

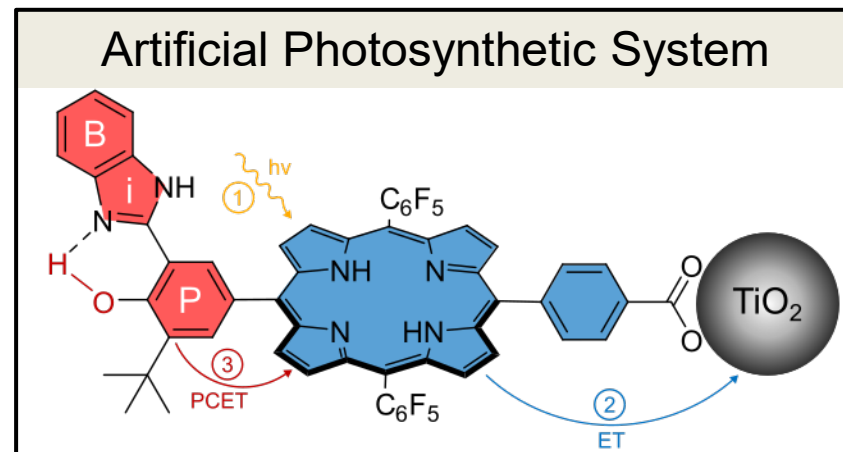
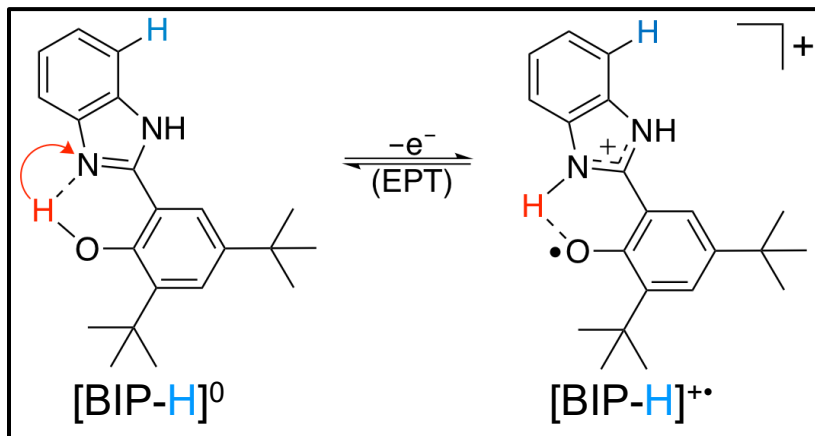
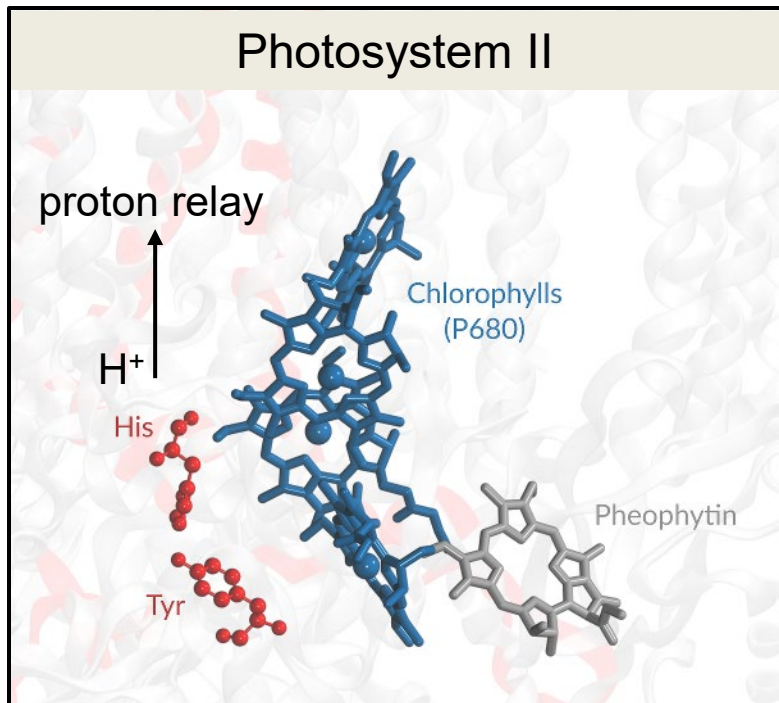
PCET in Photosystem II



Water splitting: $2\text{H}_2\text{O} + h\nu \rightarrow 2\text{H}_2 + \text{O}_2$

Can we follow nature's example and assemble a proton wire?

Artificial Photosynthetic Systems

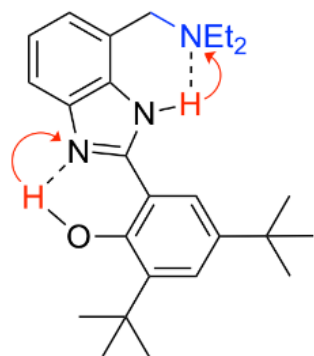


- Mimics PCET in Photosystem II for water oxidation
- Goal: design proton relay systems with desired oxidation potentials ~ 1 V vs SCE
- Calculations are guiding this design

*Mioy Huynh, Ana and Tom Moore,
Cliff Kubiak, SHS, and others*

Y. Zhao et al. PNAS 2012, 109, 15612.

Theoretical Design of E2PT Systems



BIP-CH₂NEt₂

1-electron, 1-proton transfer (EPT)
1-electron, 2-proton transfer (E2PT)

- Promote 2nd PT by increasing pK_a of 2nd acceptor site
- Predictions from theory
 - E2PT process shifts redox potential to less positive potentials by ~300 mV
 - KIE for E2PT predicted to be ~1.3, decreased from the KIE of ~2 for EPT

BIP-X	Expt $E_{1/2}^a$	Calc $E_{1/2}^a$
BIP-H	1.04 ^b	1.04 ^c
BIP-PF ₁₀	1.00 ^b	0.94
BIP-COOH	0.90	0.98
BIP-COOCH ₃	0.93	0.96
BIP-CONH ₂		0.94
BIP-CONEt ₂		0.95
BIP-CH ₂ NH ₂		0.66 ^d
BIP-CH ₂ NEt ₂		0.57 ^d

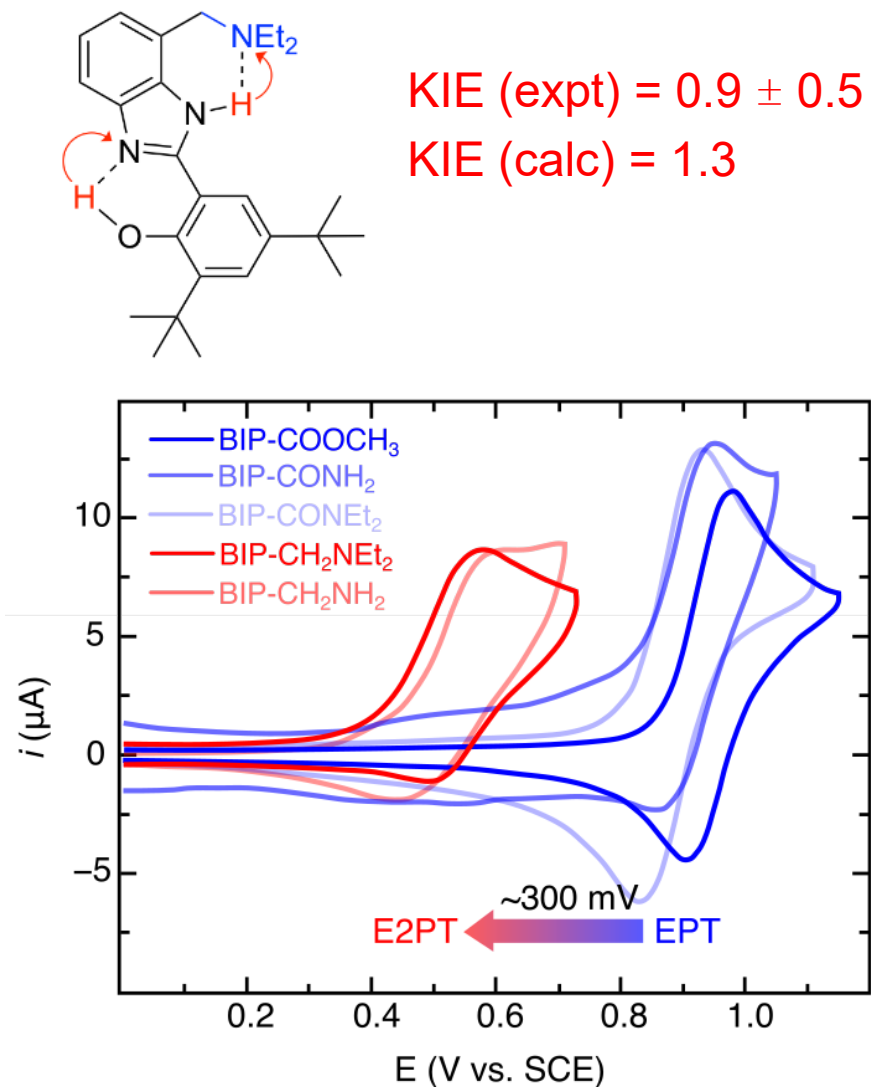
^a Units of V vs. SCE in MeCN

^b Megiatto Jr. et al., *Nat. Chem.* 6: 423 (2014)

^c Reference reaction; agrees by construction

^d E2PT product

Experimental Validation of E2PT



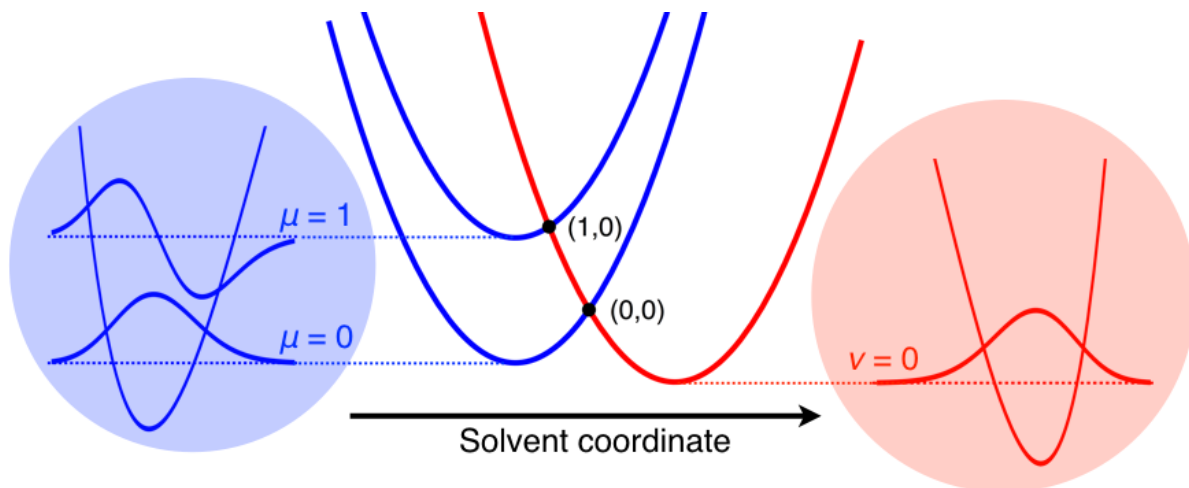
BIP-X	Expt $E_{1/2}^a$	Calc $E_{1/2}^a$
BIP-H	1.04	1.04
BIP- PF_{10}	1.00	0.94
BIP- COOH	0.90	0.98
BIP- COOCH_3	0.93	0.96
BIP- CONH_2	0.92	0.94
BIP- CONEt_2	0.92	0.95
BIP- CH_2NH_2	0.56	0.66 ^b
BIP- CH_2NEt_2	0.54	0.57 ^b

^a Units of V vs. SCE in MeCN

^b E2PT product

Huynh, Mora, Villalba, Tejeda-Ferrari, Liddell, Cherry, Teillout, Machan, Kubiak, Gust, Moore, Hammes-Schiffer, and Moore, ACS Cent. Sci. 2017.

KIEs Influenced by Excited Vibronic States

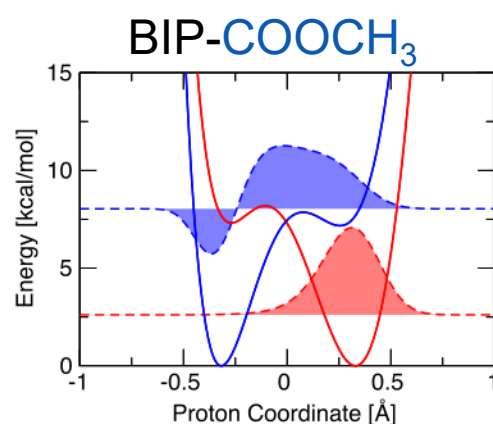
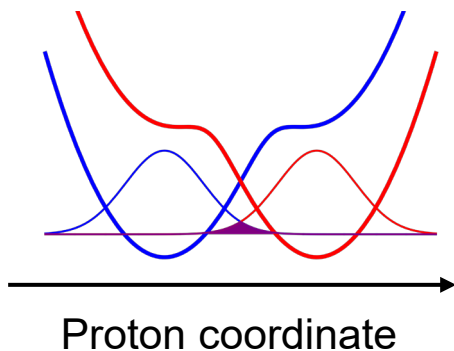


$$\text{KIE} = \frac{k_{\text{H}}}{k_{\text{D}}} \propto \frac{(\text{H overlap})^2}{(\text{D overlap})^2} \quad \text{for each pair of states}$$

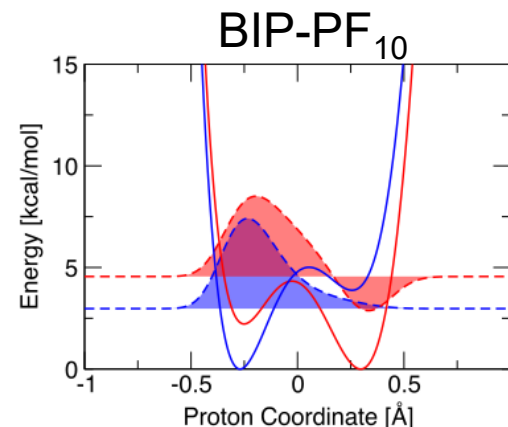
KIE increases as overlap decreases

Contributions from excited states $\rightarrow$ decrease KIE

(0,0) overlap small $\rightarrow$ large KIE
soybean lipoxygenase (80–700)



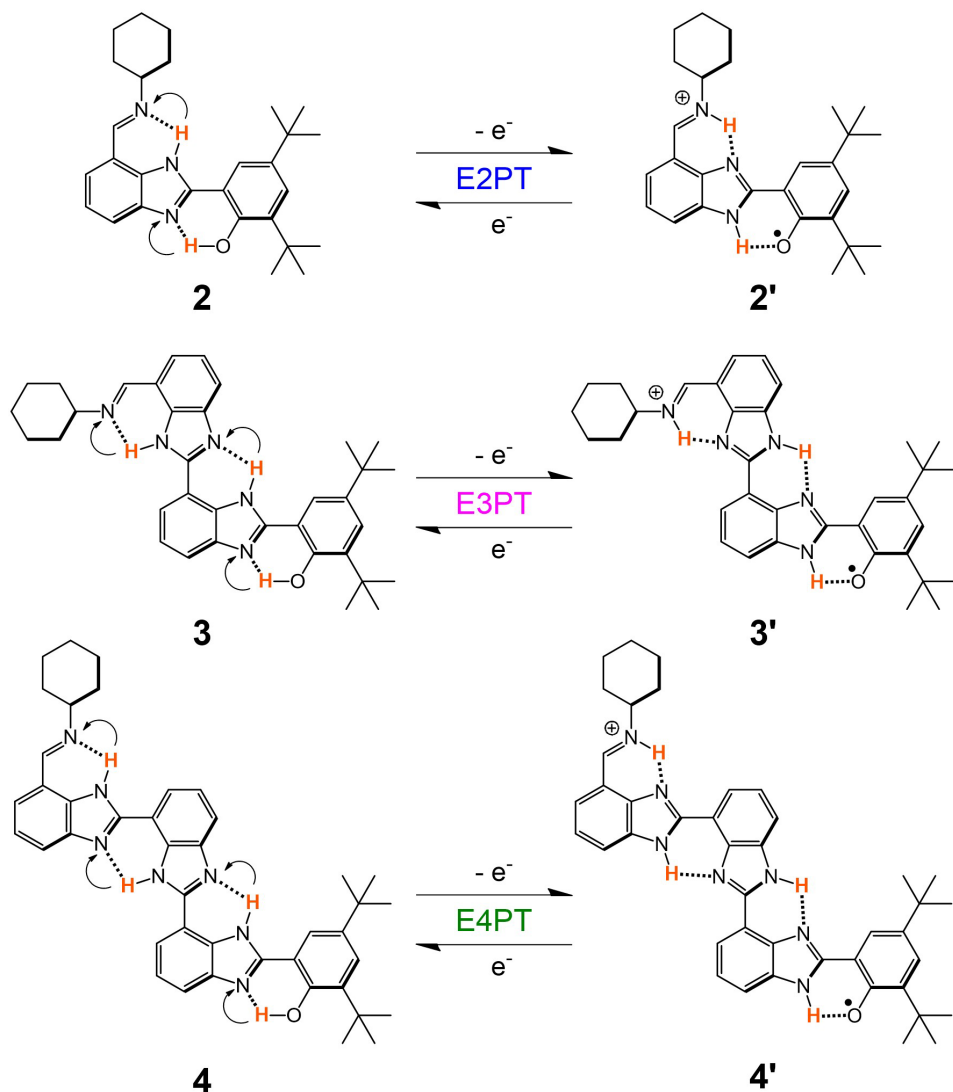
(1,0) pair dominant,
intermediate overlap
 $\rightarrow$ KIE ~ 2



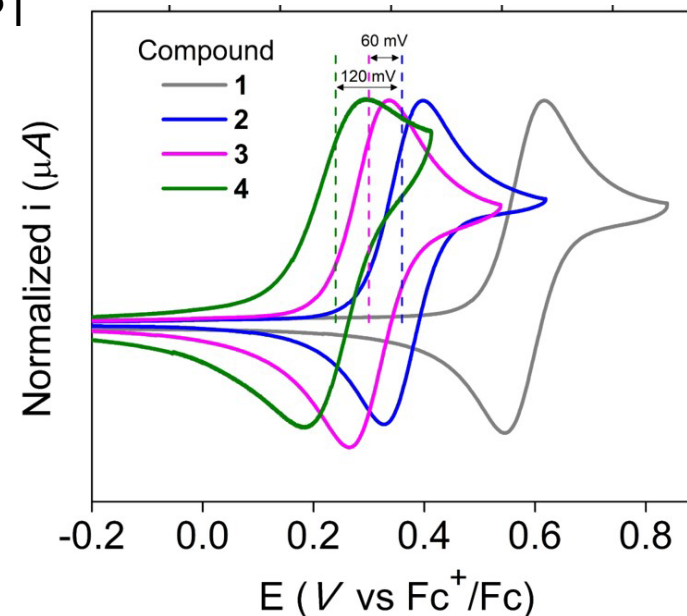
(0,1) pair dominant,
overlap nearly unity
 $\rightarrow$ KIE ~ 1

E2PT systems:
KIE ~ 1
2D proton potentials,
excited states with
large overlap

Long-Range Proton Translocation



1: E1PT

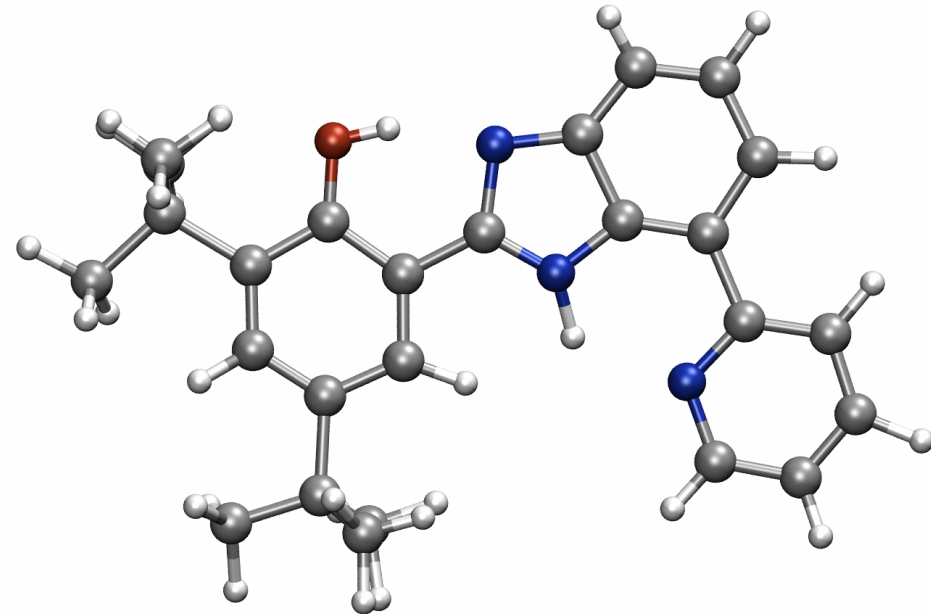
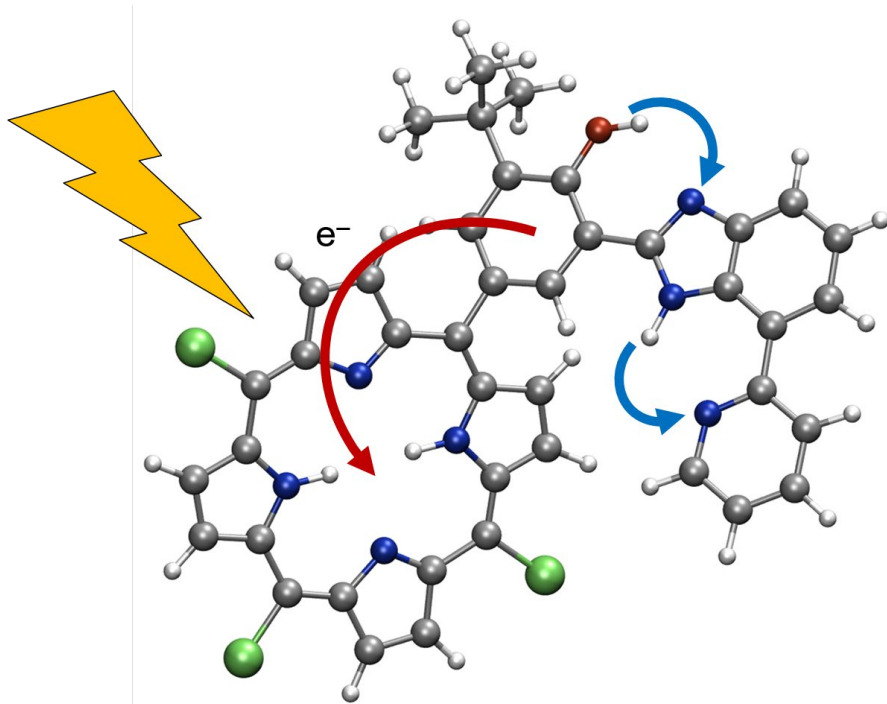


PCET can reversibly move protons up to 16 Å across H-bonded network (confirmed by IRSEC)

Odella, Wadsworth, Mora, Goings, Huynh, Gust, Moore, Moore, SHS, Moore, JACS 2019.

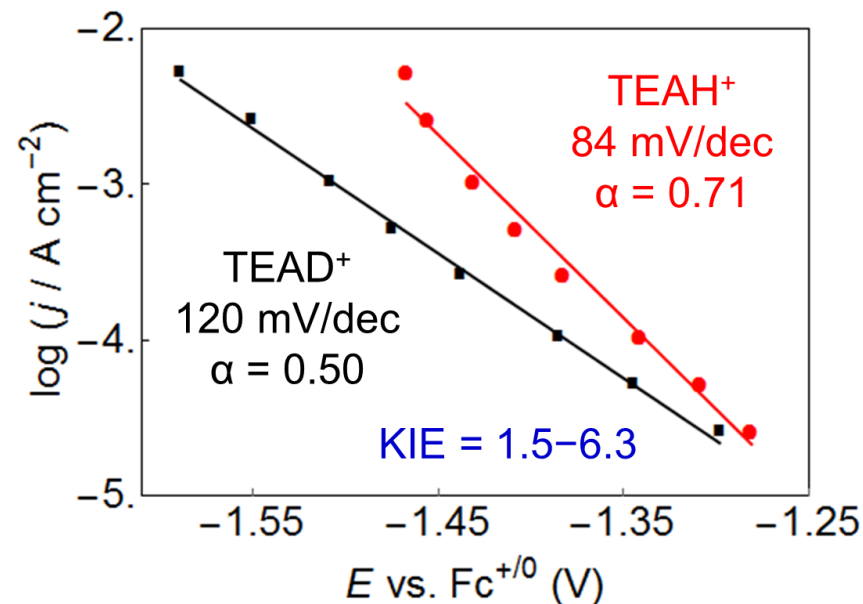
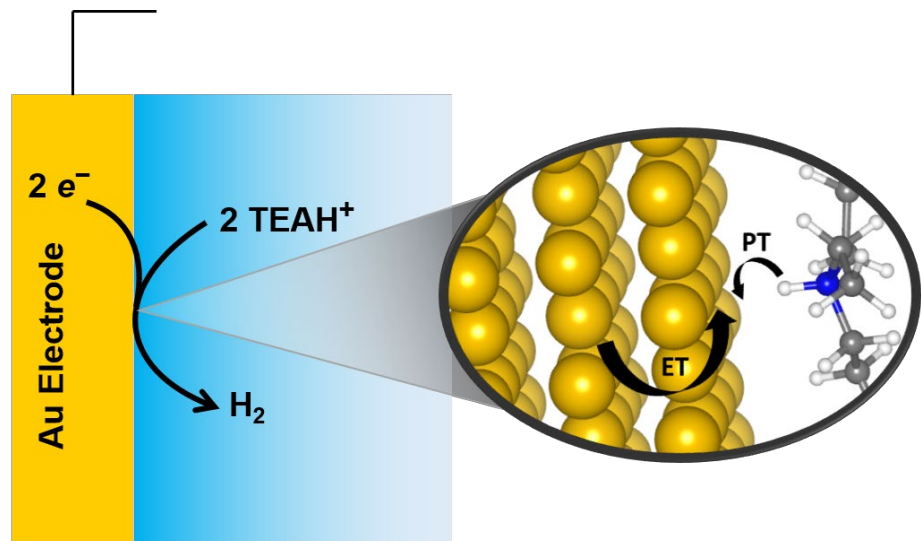
Nonequilibrium Dynamics of Photoinduced E2PT

- Equilibrate system in ground state and instantaneously remove electron
- Nonequilibrium first-principles molecular dynamics in gas phase
- Concerted but asynchronous mechanisms on femtosecond timescale
- First PT ~100 fs after ET and second PT ~600 fs later (on average)



PCET from Acid to Gold in Acetonitrile

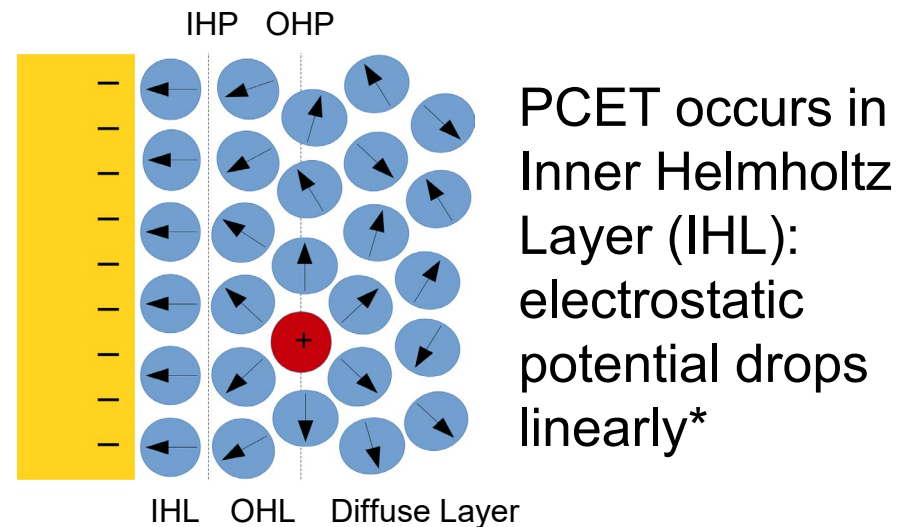
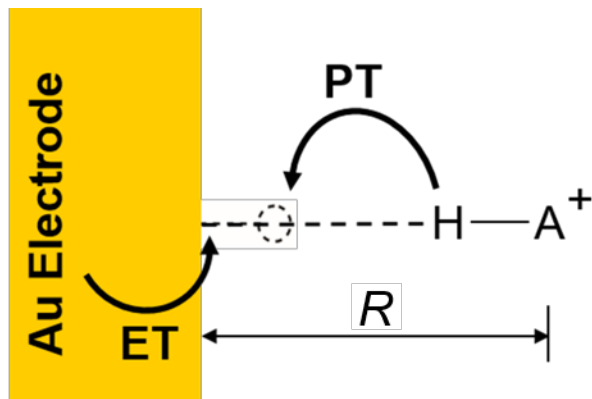
- PCET from TEAH acid to gold surface in acetonitrile
- Rate limiting for the hydrogen evolution reaction
- Experiments from Surendranath group show potential-dependent KIE
- PCET calculations reproduce these results and explain in terms of different contributing excited vibronic states for H and D, dependent on E



Experiments: Jackson, Surendranath, JACS 2016.

Theory: Goldsmith, Lam, Soudackov, Hammes-Schiffer, JACS 2019.

Rate Constant for Volmer Reaction



$$k(R, E) = \frac{\rho}{\hbar} \sqrt{\frac{\pi}{\lambda k_B T}} \sum_{\mu, \nu} P_{\mu} V_{\mu \nu} (R)^2 \int d\varepsilon f(\varepsilon) \exp \left[-\frac{(\Delta G_{\mu \nu} (R, E, \varepsilon) + \lambda)^2}{4 \lambda k_B T} \right]$$

↑
Function of distance of HA^+ from electrode (R) and electrode potential (E)

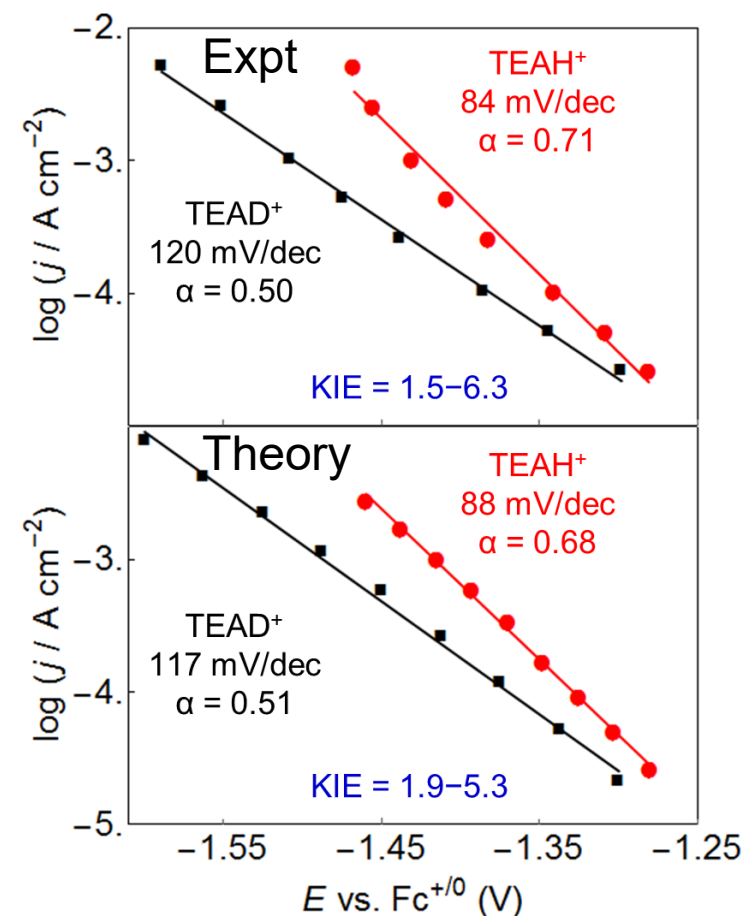
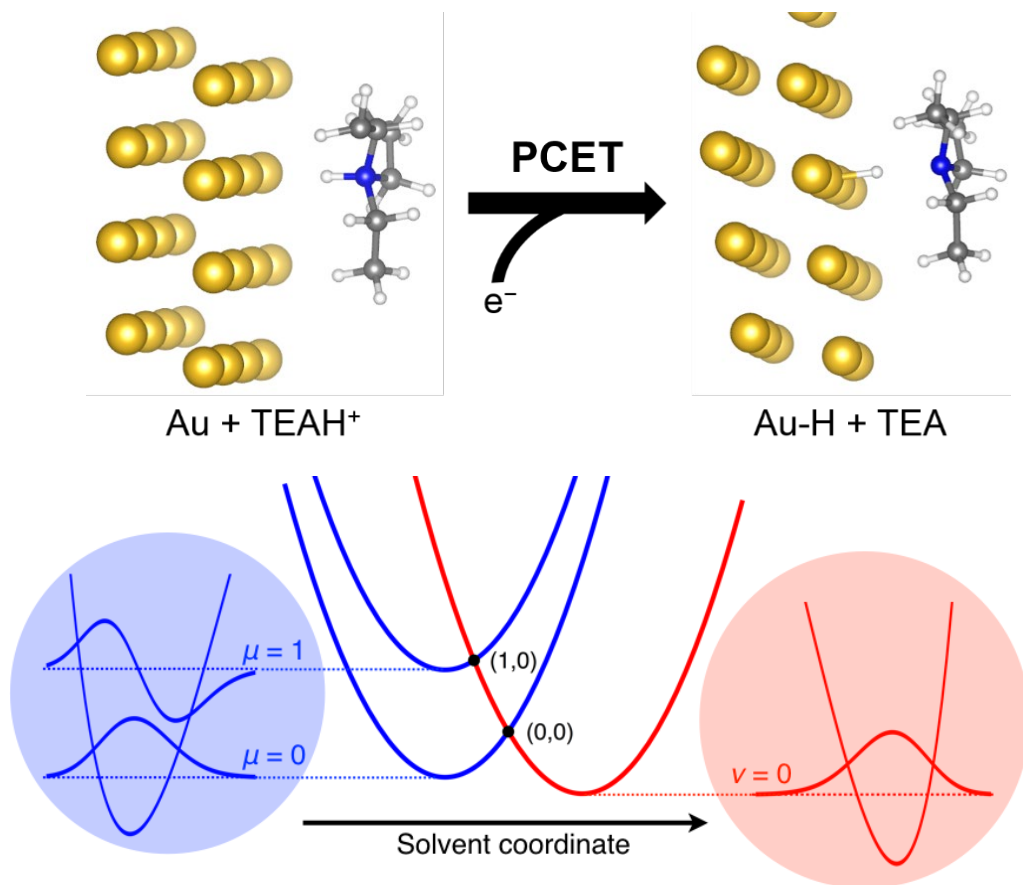
↑
Use general form of vibronic coupling that spans adiabatic/nonadiabatic limits

- $\Delta G_{\mu \nu} (R, E, \varepsilon)$: sum of free energy difference between states, $eE + \varepsilon$, and electrostatic/non-electrostatic work terms due to double layer effects
- Current density: integrate k over R , weighting by concentration of AH^+

*extended Guoy-Chapman-Stern model: dielectric continuum

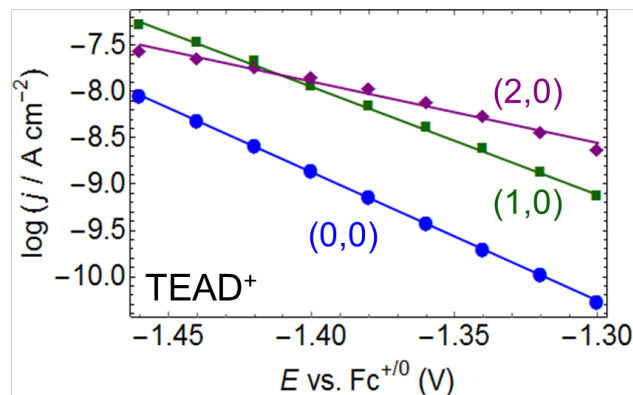
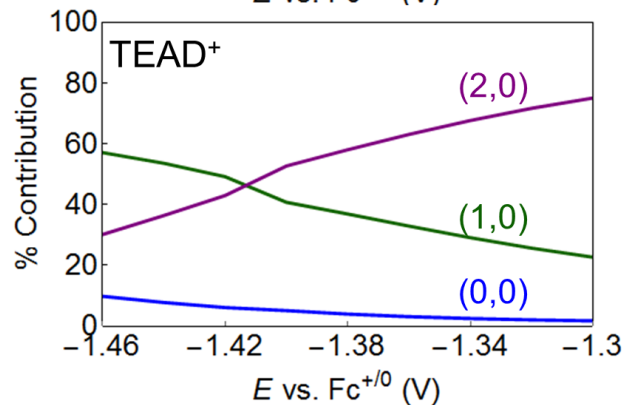
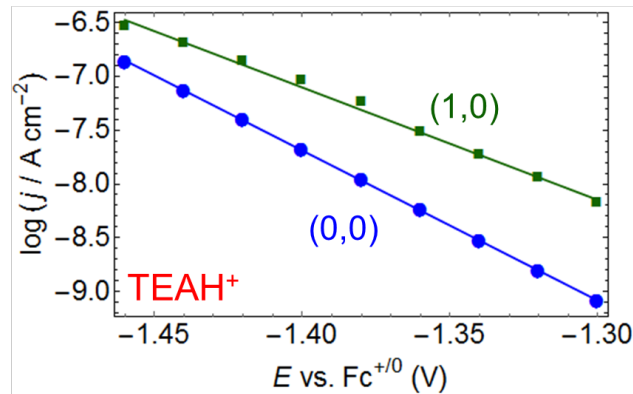
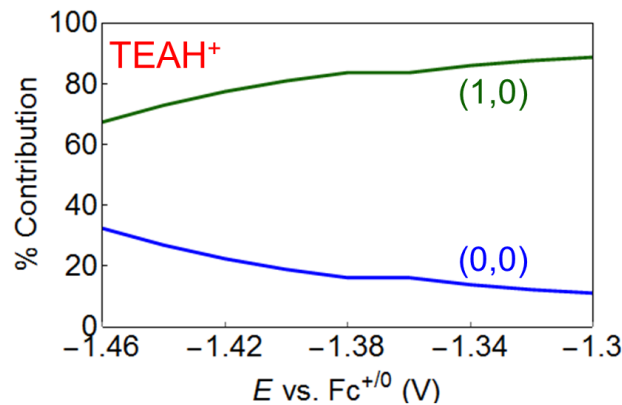
Comparison of Theory and Experiment

- Theory reproduces potential-dependent KIE (Tafel slopes and transfer coefficients α) with physically reasonable parameters
- Physical explanation: different pairs of proton vibrational states dominate for H and D with different dependence on E



Explanation for Potential-Dependent KIE

- Dominant vibronic state pairs depend on E and are different for H and D
- Smaller level splittings for D $\rightarrow$ greater contributions from excited reactant states
- Potential-dependent KIE due to different contributions for H and D

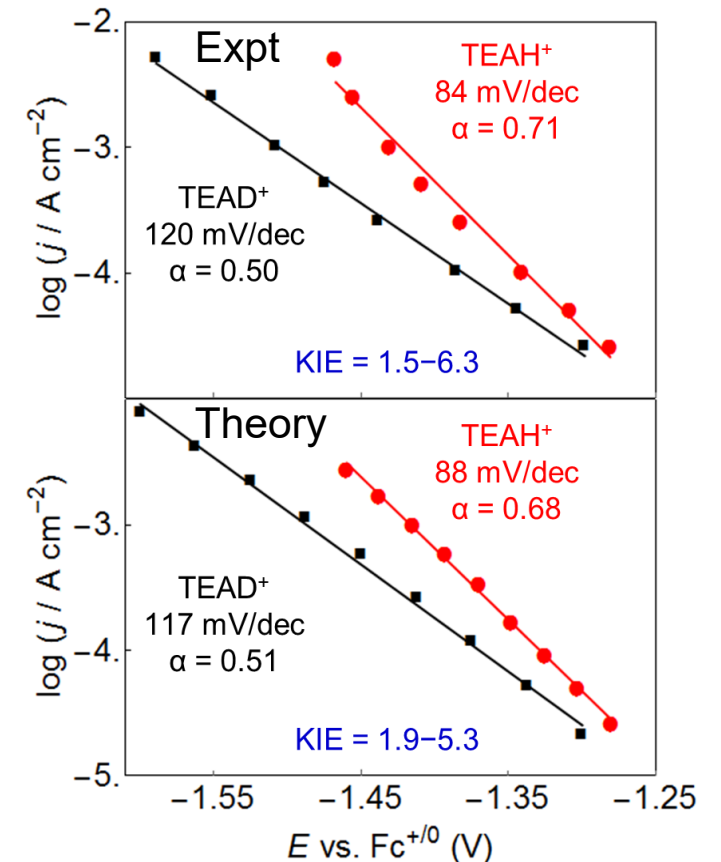
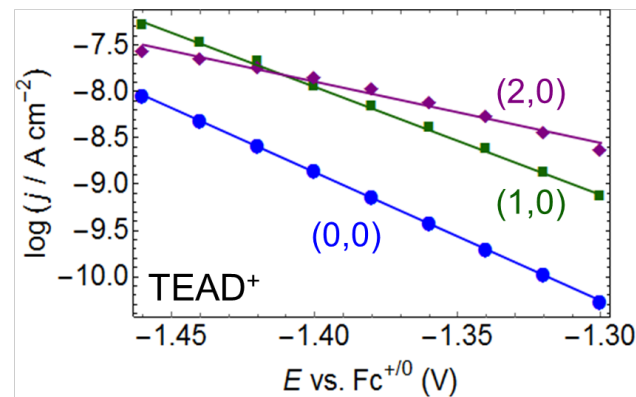
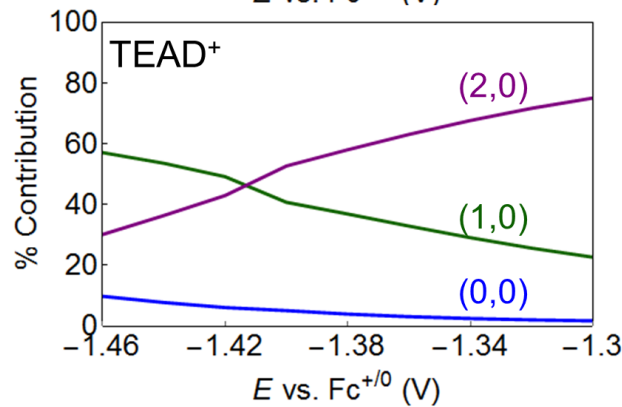
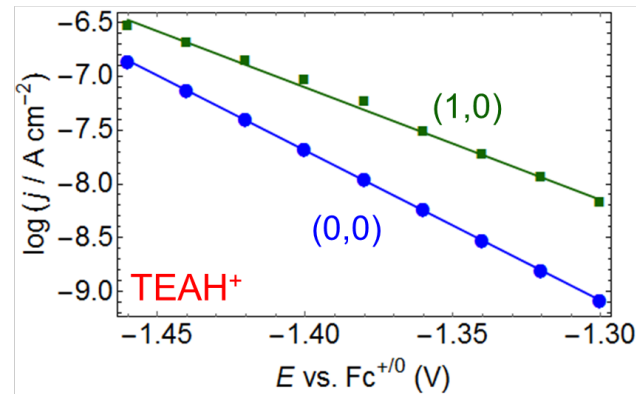
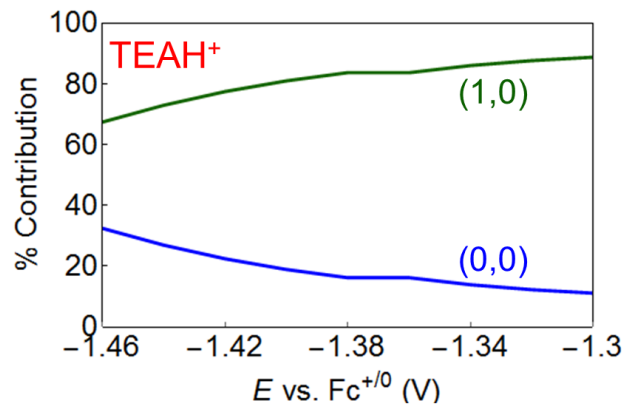


$$\alpha \sim \frac{1}{2} + \sum_{\mu, \nu} w_{\mu \nu} \left(\frac{\Delta G_{\mu \nu}^{\circ}}{2\lambda} \right)$$

percent contribution
to rate constant

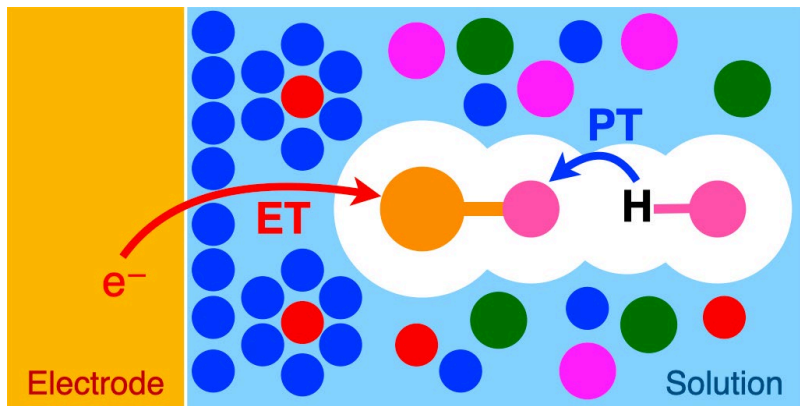
Explanation for Potential-Dependent KIE

- Dominant vibronic state pairs depend on E and are different for H and D
- Smaller level splittings for D $\rightarrow$ greater contributions from excited reactant states
- Potential-dependent KIE due to different contributions for H and D

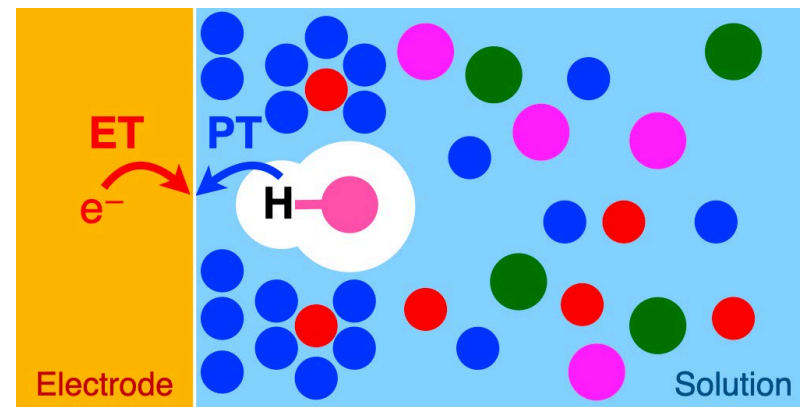


Homogeneous vs Heterogeneous PCET

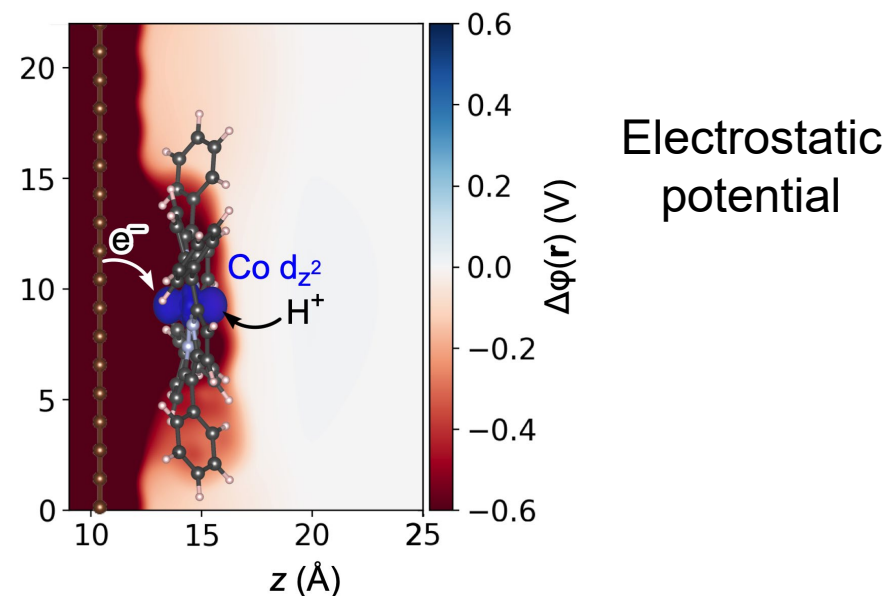
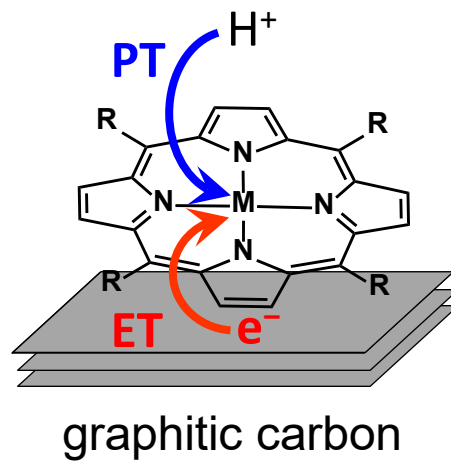
Homogeneous:
electrode does not participate in chemistry



Heterogeneous:
electrode participates in chemistry

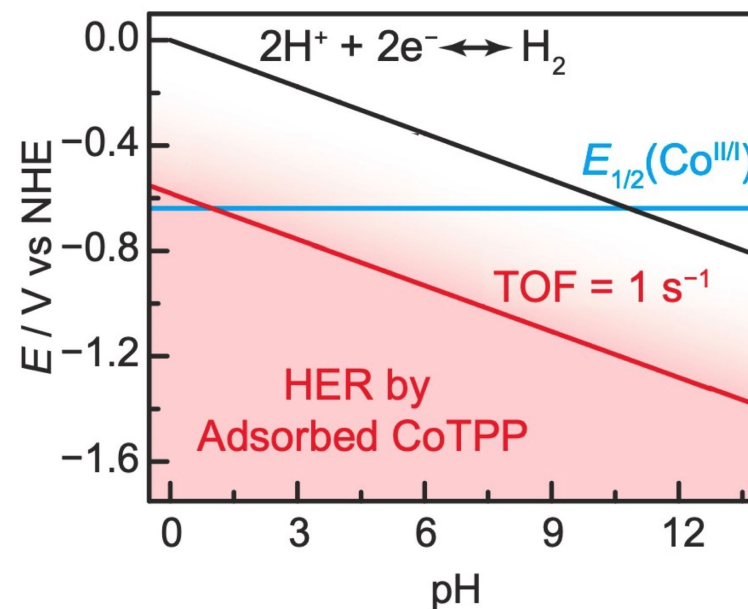
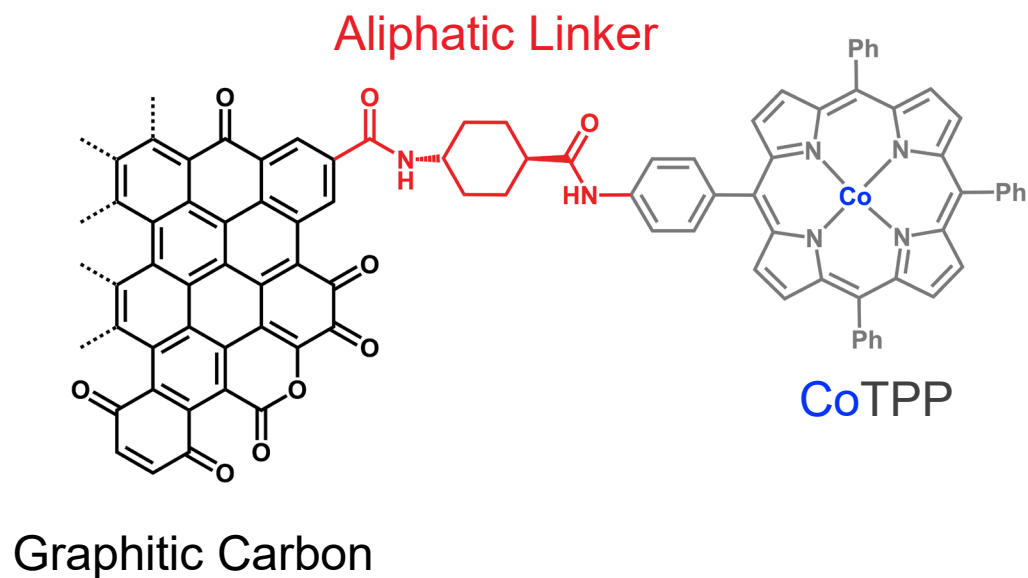
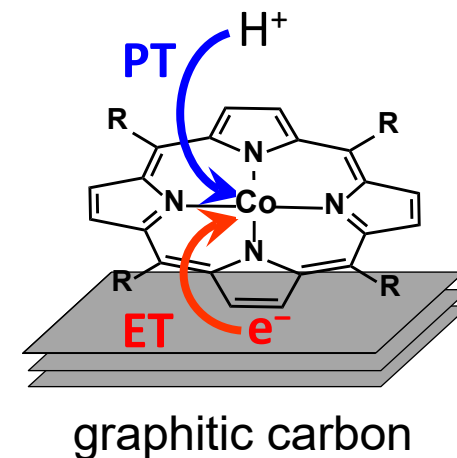


Heterogenized molecular catalyst:
electrode/molecule system
participates in chemistry



CoTPP on Graphitic Carbon in Water

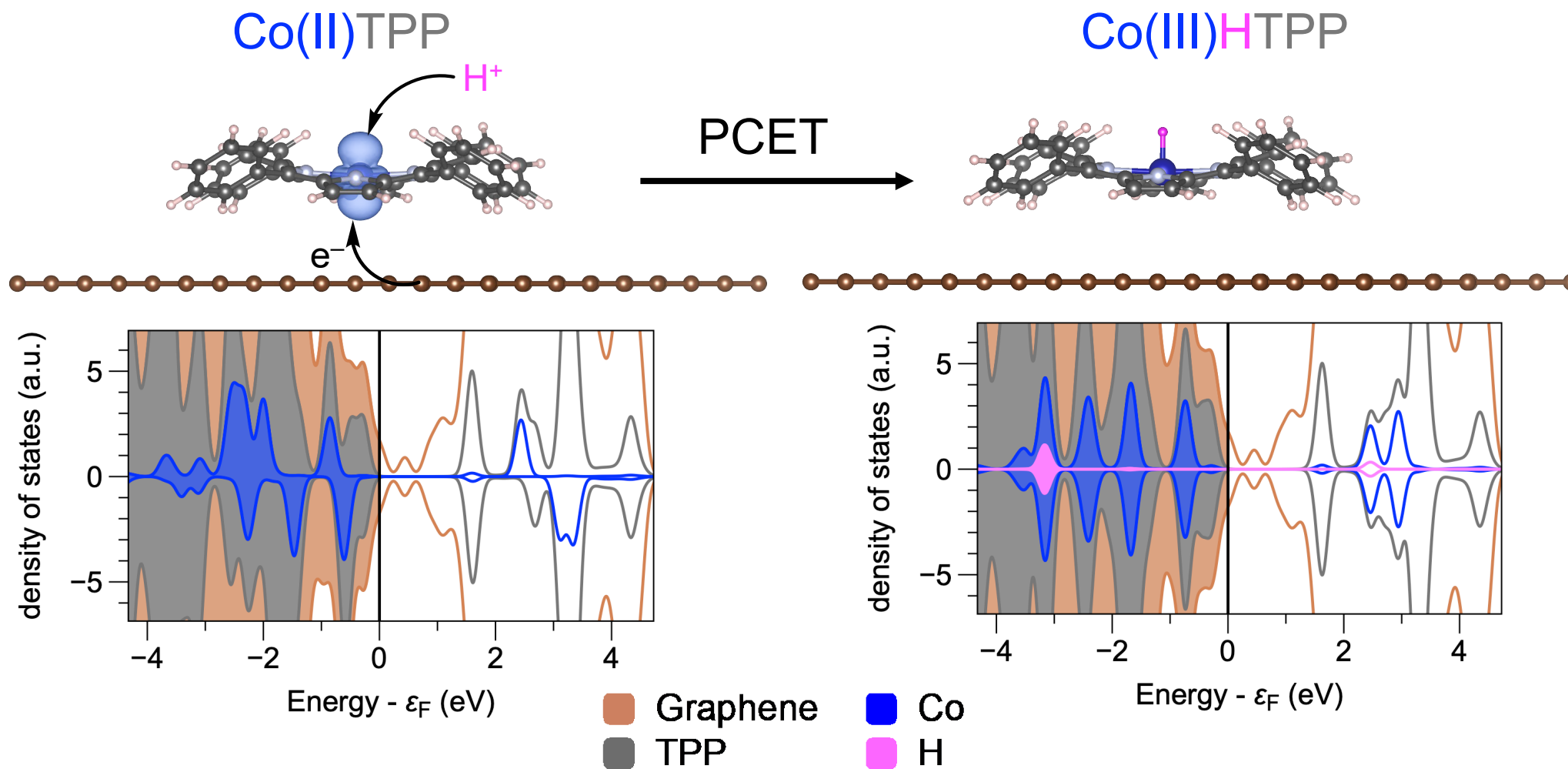
- CoTPP not soluble in water → heterogeneous PCET
- Surendranath group experiments: concerted PCET, directly to Co(III)H
- HER occurs across all pH with slope of 58 mV/pH unit



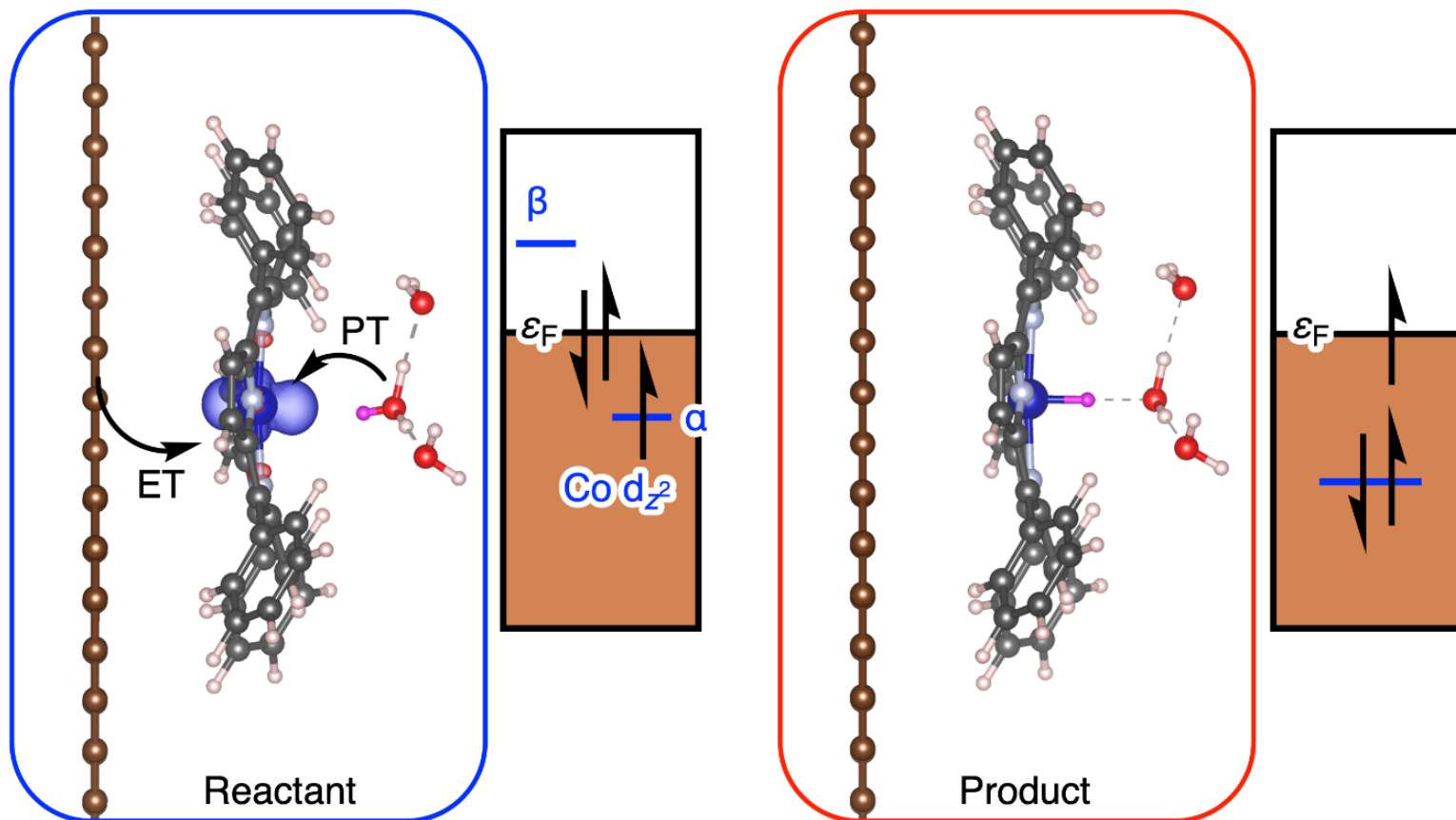
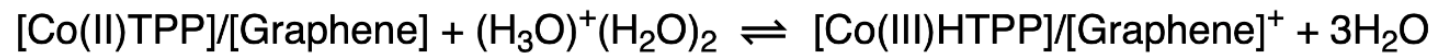
Kaminsky, Weng, Wright, Surendranath, *Nat. Catal.* 5, 430 (2022).

Hutchison, Kaminsky, Surendranath, SHS, *ACS Cent. Sci.* 2023.

PCET at CoTPP-CH is Concerted

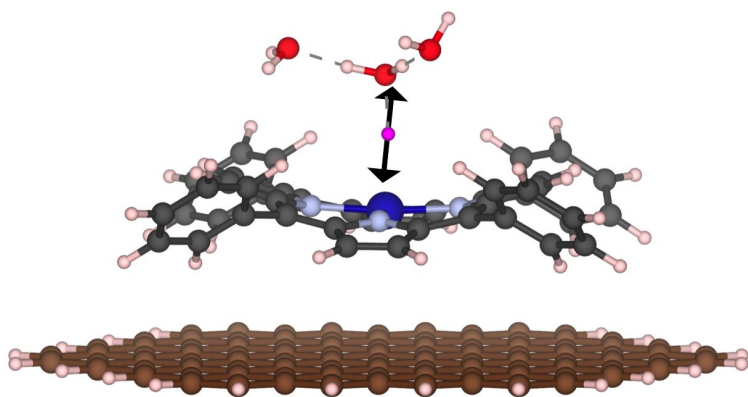


Model System for PCET at CoTPP-CH



Nonadiabatic PCET Theory for CoTPP-CH

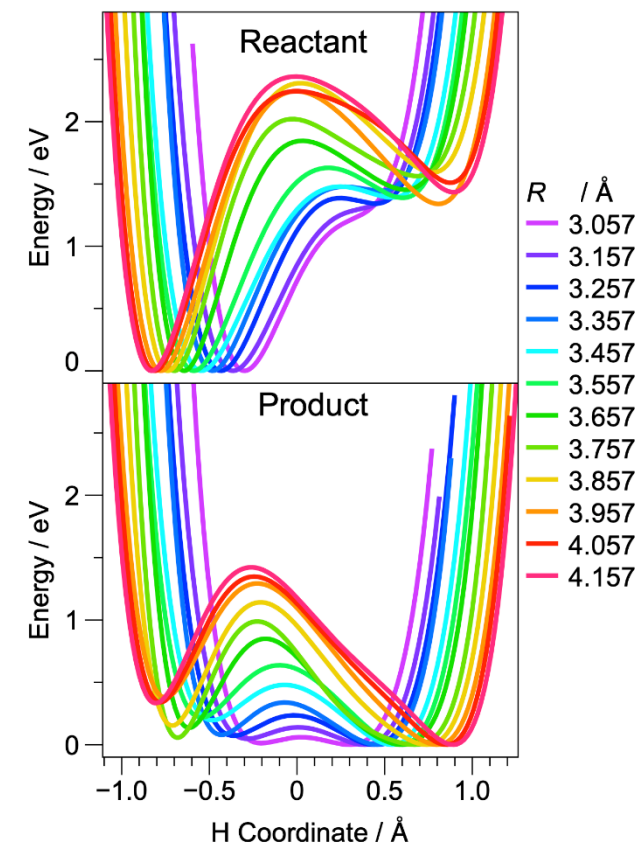
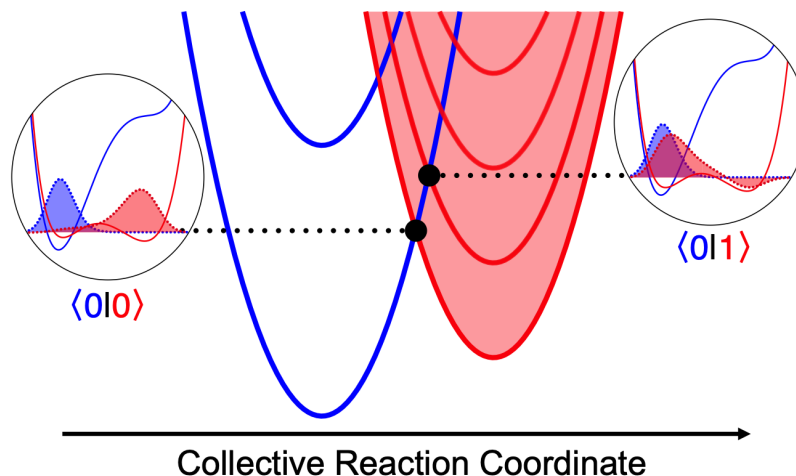
- Graphitic flake model with protonated water cluster
- Constrained DFT for each fixed proton donor-acceptor distance R to obtain proton potentials
- Main contribution to overall PCET rate constant is (0,1) vibronic state pair
- Transfer coefficients and KIEs in good agreement with experiment



Reactant: electron on graphitic flake

Product: electron on Co

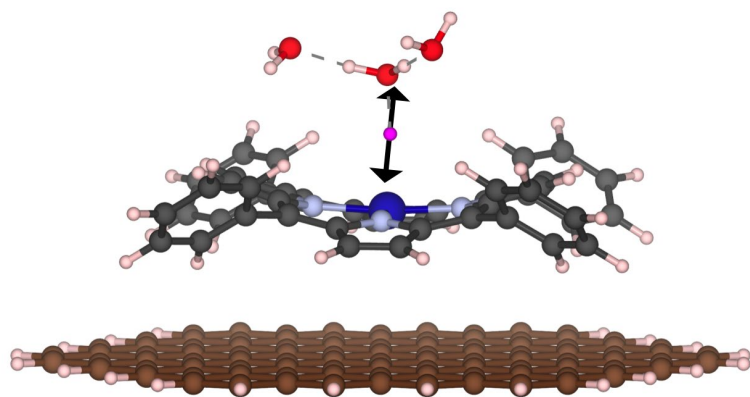
R : oxygen-cobalt distance



Hutchison, Soudackov, SHS, to be submitted.

Nonadiabatic PCET Theory for CoTPP-CH

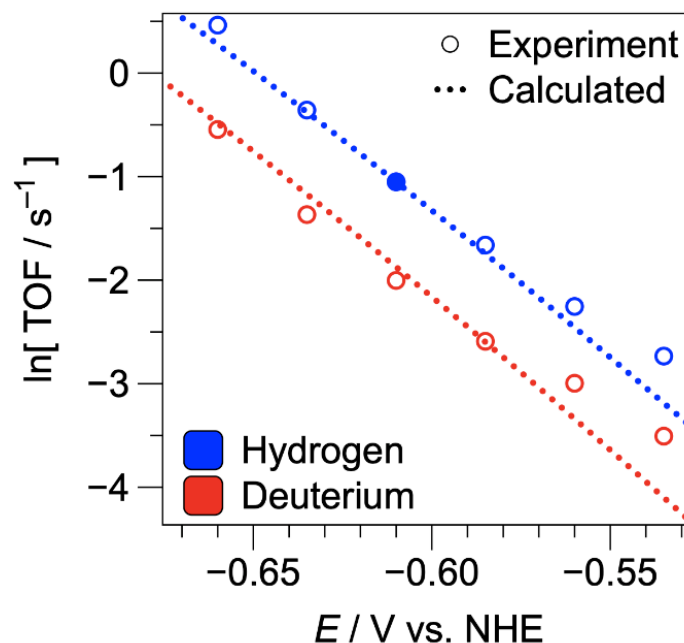
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Product: electron on Co

R : oxygen-cobalt distance

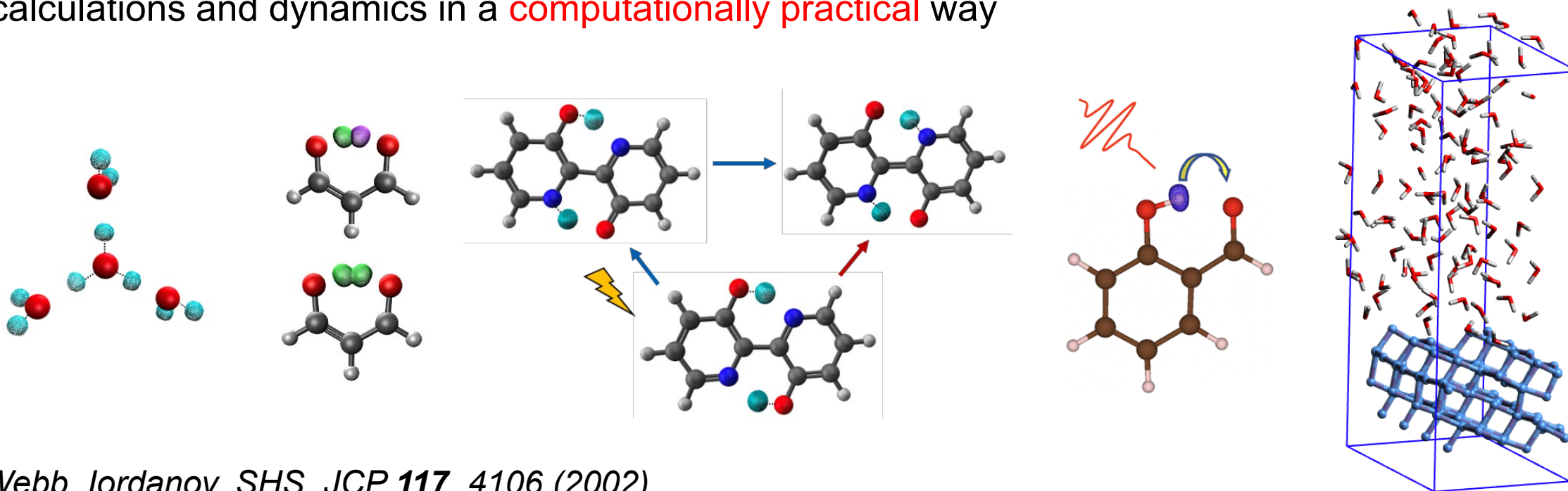


Hutchison, Soudackov, SHS, to be submitted.

Calculated rate constants scaled by constant.

Nuclear-Electronic Orbital (NEO) Method

- Treat specified nuclei quantum mechanically on same level as electrons with DFT or wavefunctions
- Solution of mixed nuclear-electronic time-independent or time-dependent Schrödinger equation
- NEO-DFT, NEO-TDDFT, NEO-CCSD, NEO-EOM-CCSD, NEO-CASSCF/MRCI, ...
- NEO includes nuclear quantum effects and non-Born-Oppenheimer effects into quantum chemistry calculations and dynamics in a **computationally practical** way



Webb, Iordanov, SHS, JCP **117**, 4106 (2002).

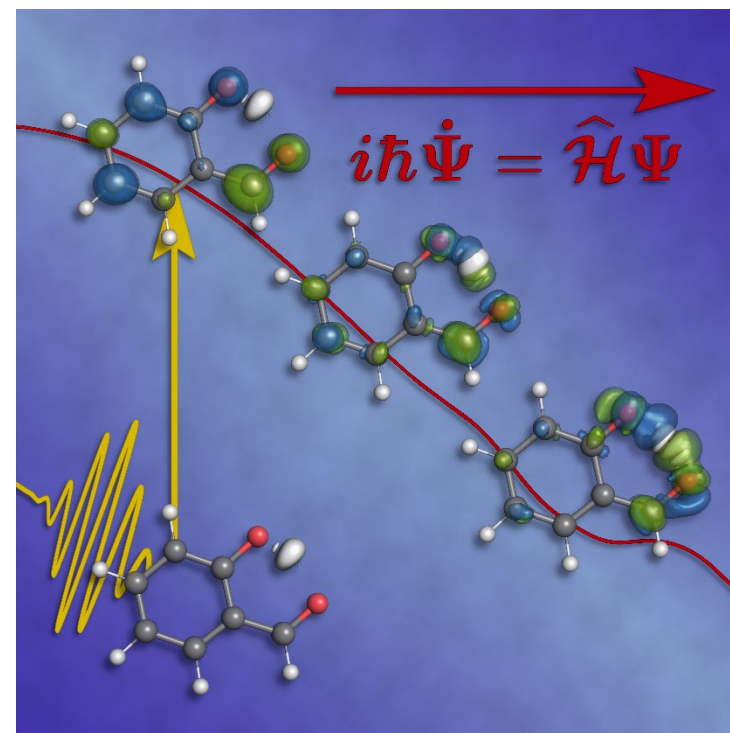
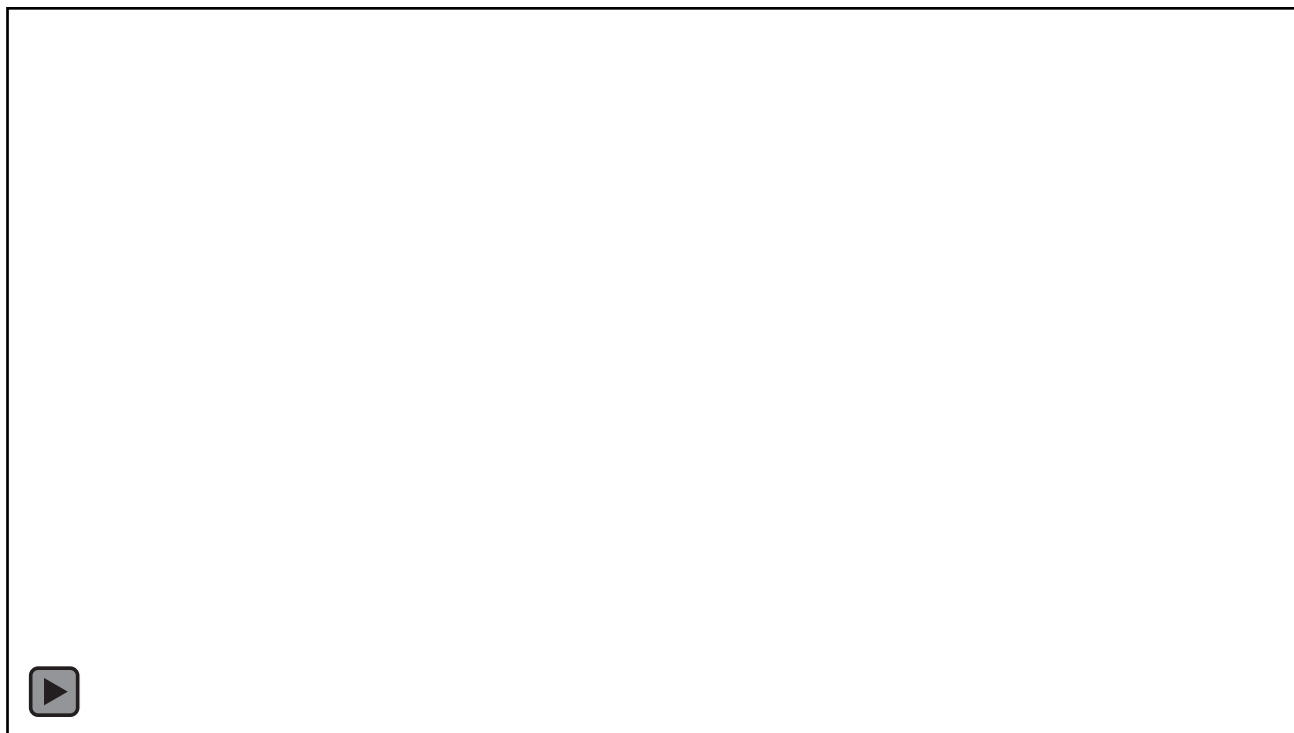
Pavošević, Culpitt, SHS, Chem. Rev. **120**, 4222 (2020).

SHS, JCP **155**, 030901 (2021).

TiO₂/water interface with
quantized protons

Photoinduced Intramolecular Proton Transfer

- Integrate nuclear and electronic dynamics in time-dependent framework
→ dynamics beyond the Born-Oppenheimer approximation
- Nonequilibrium dynamics of photoinduced proton transfer or PCET

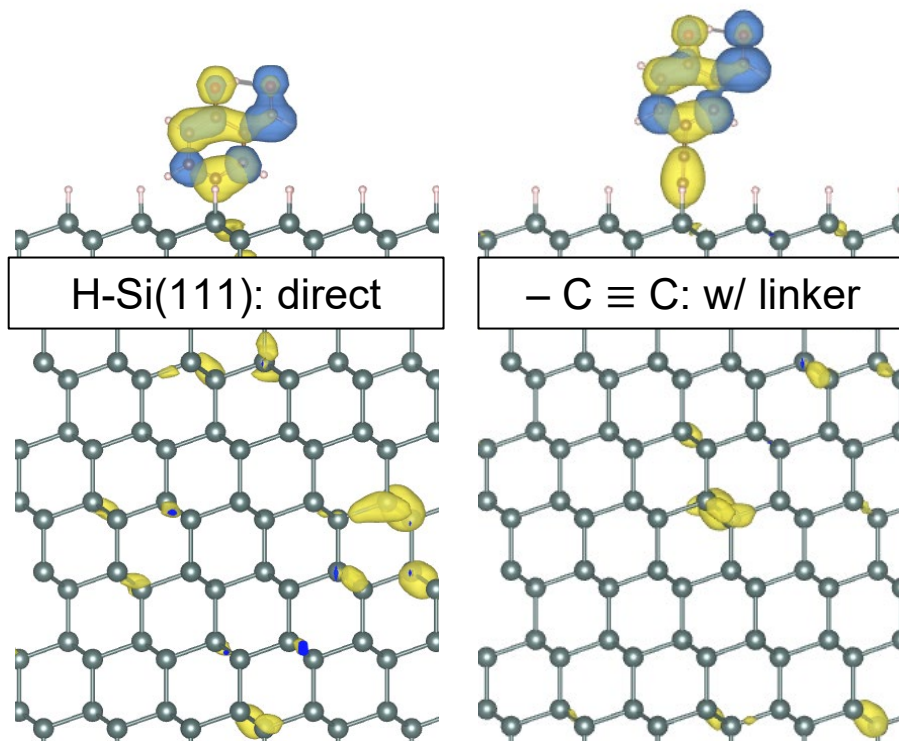


Zhao, Tao, Pavosevic, Wildman, SHS, Li, JPCL 2020.

Note: Movie shows RT-NEO-TDDFT for fixed classical nuclei

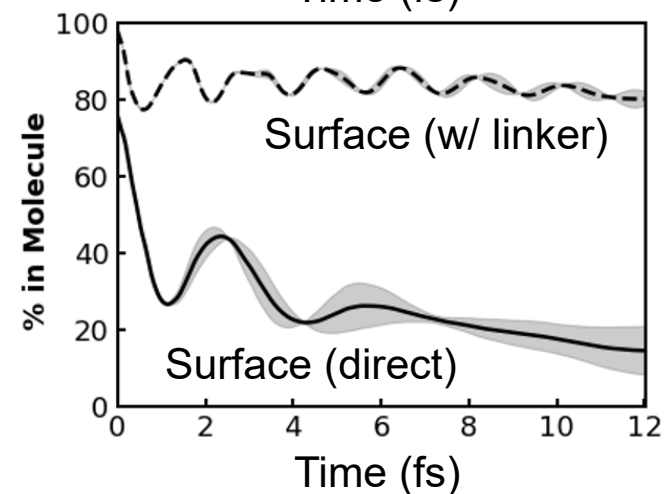
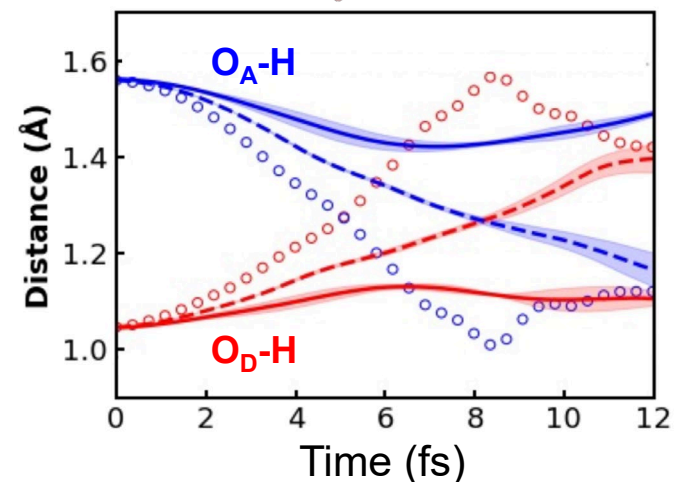
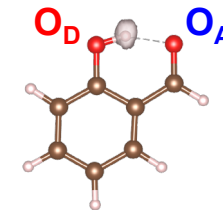
Periodic RT NEO-TDDFT

- oHBA attached to Si(111) surface directly or via linker group
- Photoexcite hyb-HOMO (yellow) to hyb-LUMO (blue)
- Linker: proton transfers, although slower than vacuum
- Direct: excited electron transfers to Si surface, suppressing PT



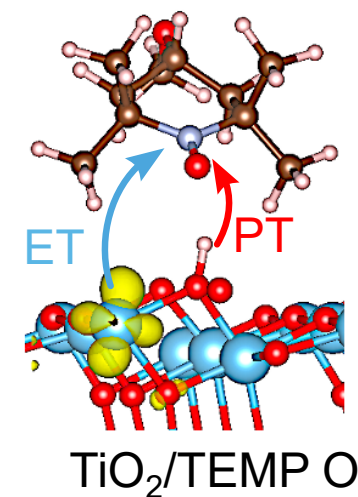
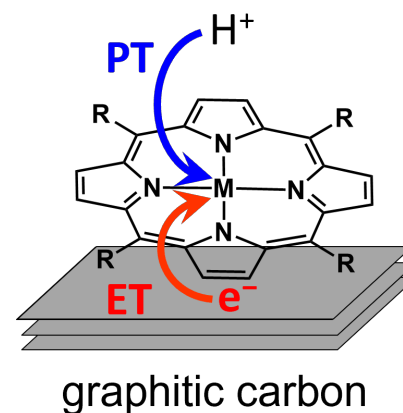
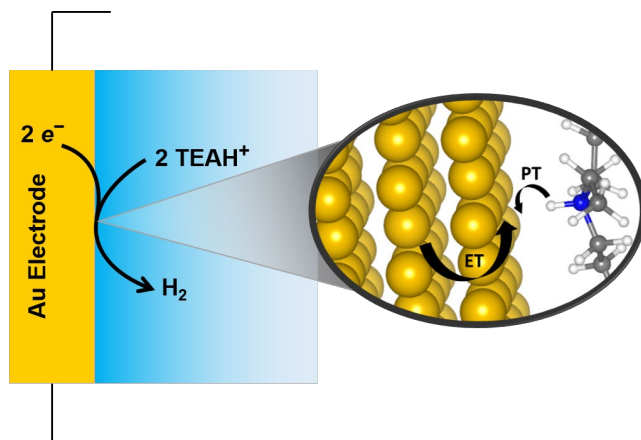
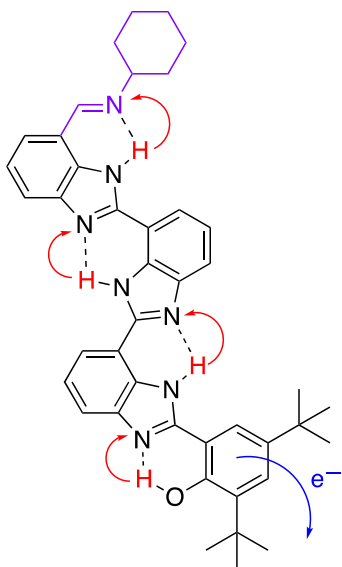
○ ○ ○ ○ Gas Phase
 — Surface (direct)
 - - - Surface (w/ linker)

% excited electron
on molecule

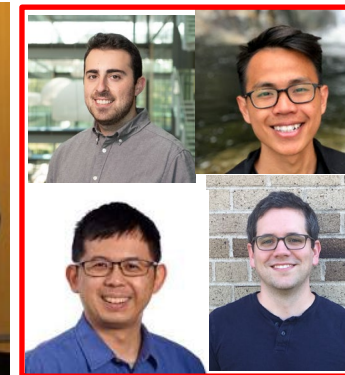


Summary

- Developed a general theory for PCET and derived rate constant expressions for PCET in solution, proteins, and electrochemistry
- Devised methods for computing input quantities to rate constant expressions
- Important aspects of PCET
 - Hydrogen tunneling and contributions from excited vibronic states
 - Tunable via substituents that impact thermodynamics
 - KIE can range from 1 to more than 100
- NEO method can be used to compute real-time nuclear-electronic dynamics and input quantities



Acknowledgments



Alexander Soudackov, Mioy Huynh, Josh Goings, Zachary Goldsmith, Yan Choi Lam, Phillips Hutchison

Experimental collaborators: Tom Moore, Ana Moore, Yogi Surendranath

Theory collaborators: Xiaosong Li (RT-NEO), Yosuke Kanai, Volker Blum (periodic NEO-DFT and TDDFT)

Funding: NSF, DOE

NEO method: GAMESS, Q-Chem, Chronus Quantum, FHI-aims